

Manual

Shop Floor / Order Data Management BDE-BDM 8.1

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1 Overview Shop Floor / Order Data Management

Purpose

The function package “shop floor / order data management” provides an extensive range functions for recording times and quantities in orders / operations. In addition, it provides the ability to correct entered data later on.

Implementation considerations

You use the function package if you:

- would like to enter times and quantities for orders / operations so that these can be posted back to other systems, if required
- would like to modify data entered in the system at a later date.

Features

- Order administration
 - Configuration used to set customer-specific order numbers, order sequence numbers, operation numbers or split number lengths
 - Support for different types of orders like production orders, overhead cost orders, capacity orders, maintenance orders and project orders
 - Administrative function for locking or releasing operations
 - Administrative function for terminating operations that are not (no longer) logged on.
 - Administrative function for reactivating inadvertently logged off operations
- Order-related entry and posting functions:
 - Log-on /off, interrupt and partially confirm orders or operations
 - Entering yields, scrap quantities, rework quantities or open quantities
 - Posting of recorded production and downtimes to freely definable resource performance accounts
 - Comparing recorded workplace / machine times with the shift model stored at the workplace
 - Posting runtimes and numbers of units per operation
- Maintaining entered data using posting maintenance:
 - Displaying order-related log records (postings) generated based on the data entered
 - The ability to edit and correct order-related log records
 - Generating cancellation log records during processing after uploading to the ERP system
- Event maintenance
 - The ability to display and correct entered BDE events
 - The ability to edit and correct BDE events as an alternative to posting maintenance

- Recalculation function based on changed/corrected BDE events Generating updated log records (postings)
- Generating cancellation log records during processing after uploading to the ERP system

2 Fields of Application for HYDRA-BDE

2.1 Summary

The goal of the following description is to present the many collection and processing options offered by HYDRA shop floor data collection (HYDRA-BDE) and machine data collection (HYDRA-MDE).

2.2 Basics of shop floor data collection

The postings (e.g. logon/logoff, end/interruption, etc.) associated with shop floor data collection in the HYDRA system are generally based on the **machine**, **order** and **person** components. Depending on the approach and the main focus of the individual use of HYDRA, different components might play a prominent role or they might be unnecessary and therefore as far as possible disabled.

Machine/workstation – where does production happen?

Logging an order or employee on or off is basically machine or workstation related. Logging on to a machine answers the question of where a BDE posting takes place. BDE postings can basically be executed only on those machines or workstations which have been configured accordingly in the HYDRA master data.

The designations *machine* and *workstation* are synonymous in HYDRA. In HYDRA-MDE, the expression "machine" is favored. In HYDRA-BDE, the designation "workstation" is favored.

Order/operation - what is produced?

All tasks which employees perform on machines are order related. Logging the order or its operation on answers the question what is done or which activity is executed.

HYDRA differentiates between different order types to classify orders according to their utilization. Different control information that is decisive for managing the orders is defined for each order type.

HYDRA provides the following order types by default. Further order types can be created according to the customer's requirements:

- Production order

A production order generally relates to the article/item and is characterized by a target quantity as well as a completion date.

- Overhead cost order

An overhead cost order (e.g. cleaning of the workplace) normally only refers to cost-accounting.

- Capacity order

A capacity order has only been designed for planning purposes (assignment of capacities) within HYDRA shop floor scheduling. Normally, capacity orders are later converted into production orders.

- Project order

A project normally is unique. The project steps to be performed as a part of the project can be summarized in a project order.

- Maintenance order

A maintenance order has been designed for the planning or recording of maintenance measures.

In general, orders are created in a higher level system (e.g. ERP system) and transferred to HYDRA using an interface. Nevertheless, orders can also be created manually in HYDRA. Planning of orders (in the ERP system or HYDRA) depends on precise requirements.

Orders are mostly multi-level and divided into several operations to be processed on different workstations and machines. The order and operation are generally logged on to the shop floor terminal at the individual workplace. Data can only be posted onto operations at shop floor terminals if a corresponding pool of orders exists in the HYDRA database.

The terms 'order' and 'operation' are mostly synonymous in the handbook descriptions. Usually, when posting functions on the terminal are being described, the operation is meant, and documentation on HYDRA shop floor data collection also emphasizes operation data. Only in certain HYDRA descriptions is the 'Order' used as an umbrella term for the whole multi-level production order.

Person - who is working?

Logon and logoff of the employee to/from the shop floor terminal is used in HYDRA to create the relation to the machine and operation. The work confirmations of employees form a basis for the calculation of personal expenses and for performance determination. The logon/logoff of personnel answers the question, who is working where and what they are working on.

The relation of a posting to an employee is mostly optional. The logging on and off of production personnel is subject to fundamental decisions which are made before an MES system is introduced. HYDRA usually only allows those employees to log on, who are stored in the HR master data of the HYDRA database and authorized in the company.

2.3 Presentation of recording results

The components of data collection - machine, order, person, - are summarized to the greatest possible extent in logical dialogs. Consequently, an employee may, for example, log on and off to and from a terminal together with an order.

Decisions as to the design of individual dialogs on the shop floor terminals depend on the environment in which it is used, the industry sector, the organization, the range of items, the order structure and the machinery involved.

The data collection options in HYDRA represent a range of exemplary uses, which are supported by the modules of the standard system. A single installation requires only one selection of the possible configuration options. Individual HYDRA settings are finalized by the HYDRA user, usually together with an MPDV consultant, after successful installation and in the course of customizing.

2.3.1 Time Recording

"Posting events" on shop floor terminals form the basis for the posting of particular machine times to an operation. A time posting is initiated by the posting event "logon" and completed logically by the posting event "logoff" or "interruption". This basic principle applies to all order and personnel posting events.

The time posting of shop floor data collection primarily occurs in two different time accounts, one for machine scheduling duration and the other for labor utilization.

The **machine duration** is determined here by the time interval between logon and logoff of an operation. The machine scheduling time is compared with the shift calendar and planned shift breaks are removed from the time interval calculation.

Labor utilization represents the total of all labor times for each operation. This period is determined by the time interval between logon and logoff of the user or users. The basis for personnel postings is, once again, the shift model of the workstation and the breaks it contains. If employees are processing more than one order in parallel, then the HYDRA system carries out a proportional calculation of the operating time of each operation.

For further details on this please refer to the section entitled "posting of times".

The results are documented in "log records" that are generated automatically due to posting events.

2.3.2 Quantity collection

Quantities are separately maintained in HYDRA for each data collection level: machine, order, and person:

- **Machine related**
e.g. for display of the shift performance in the machine efficiency report
- **Operation related**
e.g. for display of the target-actual comparison in the order information
- **Personnel related**
e.g. as the basis for incentive pay based on quantities

Depending on the data collection type, quantities are either entered **automatically** (MDE/BDE terminal with counter connection) or must be **entered manually**.

Manual quantity postings are carried out in the entry dialog at the end of personnel or order processing or as partial confirmation/upload during order processing. Depending on the objective, the entry dialogs *Interrupt operation*, *Logoff operation*, *Logoff person* and *Partial confirmation/upload* are configured for the input of quantities. Quantities that are recorded manually are always posted onto the operation that is logged on. Posting onto the person realizing the posting depends on the posting type or the respective parameter settings configured in HYDRA.

Along with manual entry of quantities, HYDRA also supports **automatic entry of quantities** with active machine data collection. With this type of entry of quantities, a continuous quantity posting to the machine, and to the logged on operation and the logged on employees takes place at cyclical intervals.

All OPs and employees logged on to the machine are automatically assigned quantities. The quantities are computed according to the partitioning/cavity that is respectively defined for the operation and are posted onto the operations or persons.

HYDRA provides different accounts for quantity posting. The accounts that are predominantly used are yield and scrap. The two fields are provided appropriately during data collection. Scrap quantities can be classified by a scrap reason. The available scrap reasons are configured in HYDRA.

"Rework" and "open quantity" are other quantity accounts. Normally, they are only used specifically for customer-specific scenarios. The fields can be made available by way of customizing.

2.3.3 Options of Machine Data Collection

By connecting a machine to the shop floor terminal an automated collection becomes possible:

- Collection of production times and downtimes
- Direct assignment of malfunction reasons by receiving operating signals from the machine control
- Piece number recording for yield, total quantity and scrap

Every workstation and machine is assigned a shift model by the HYDRA configuration. The shift model determines the capacity of the machine and gives the shift calendar a reference time for the performance of a machine. All machines for which recording of standstills is carried out in HYDRA, have a definite machine status at all times during the shift. The total of all production and downtimes during a shift gives exactly the shift standard time.

Definition of downtime reasons

The determination of the relevant machine statuses forms the basis of machine monitoring by means of machine signals. Together with the "production" status all data collection relevant downtime reasons for the machine are created in HYDRA. The user may configure them individually.

Classification in HYDRA resource performance accounts

HYDRA resource performance accounts are a system of time accounts, consisting of 12 accounts. They group together similar downtime reasons in a single account (e.g. all technical disturbances go in the "disturbance-caused interruptions" (DCI) resource performance account). During data collection, accumulating times are posted to the resource performance account to which the current machine status is assigned in the system configuration.

The standard definition of the HYDRA resource performance accounts is as follows:

No.	Abbreviation	Description
1	SUT	Secondary utilization time
2	DCI	Disturbance-caused interruption (= technical interruption)
3	LCI	Logistics-caused interruption (= organizational interruption)
4	SCI	Staff caused (personnel related) <i>interruption</i>
5	IMN	Idle mode, not scheduled (e.g. repairs)
6	IMS	Idle mode, scheduled (e.g. maintenance)
7	SET	Setup
8	STA	Startup
9	U8	Free for user (e.g. trial production)
10	U9	Free for user
11	MUT	Main utilization time; "Production"
12	FB	Neutral times, e.g. free, breaks etc.

If recording of downtimes is not required or if the recording of downtime reasons gives no clear result (e.g. for HYDRA group workplaces), then time recording generally takes place with *Production* status and is represented in the resource performance account 11 (MUT) main utilization time.

The HYDRA resource performance accounts are also kept relating to operations and persons. Machine scheduling times (machine duration) for operations and personnel result from the following calculation:

Scheduling time = total (RPA 1 .. RPA 11).

Automatic recording of number of pieces using machine signal connection

Recording a cyclical signal from the machine enables the recognition of downtimes, and by counting the recorded cycles it is also possible to record the number of pieces produced.

To ensure an accurate determination of the number of pieces, HYDRA supports the multiplication of recorded cycles with multiple accesses per cycle. "Partitioning" (also called cavity) is a tool specific parameter, which is transferred to the terminal when an operation is logged on. As a default value specific to articles/items, partitioning is defined in every operation for processing at the corresponding machine.

Automatically recorded data are posted as machine performance in HYDRA and simultaneously assigned to the operation and personnel currently logged on.

Entry	operation-related	person-related
actual quantities	X	X
machine times	X	X
labor utilization	X	X

3 Erfassungsarten der BDE (arbeitsgangbezogen)

3.1 Serial production

The serial production is the "classic" approach of manufacturing operations. This means that only one operation is produced at a machine at a time.

Serial production has the advantage that machine efficiency can be easily planned.

3.2 Parallel production

Parallel production means that more than one operation is processed simultaneously on the same machine.

At **individual workplaces** the system assumes that the two operations belong together (e.g. production of upper and lower parts with the same tool as two separated operations). Consequently, the system connects each person (who is logged on) with every operation. However, the labor utilization relating to operations is thereby halved.

At **group workplaces**, in contrast, parallel production is possible without the system linking the person to all operations that are currently being processed. This is mainly used at assembly workstations where several workers at the "Assembly" workstation process several operations in parallel. A disadvantage of this option is that the planning of such workstations is difficult.

3.3 Merged operations

This is a special type of serial production. At the planning stage (e.g. HYDRA shop floor scheduling) or at the shop floor terminal, different short running operations are grouped together in logical blocks with an easy to handle running time (i.e. merged operations). For these merged operations HYDRA creates a "representative" operation, which is logged on for all individual operations included. The entered data is divided according to different configurable perspectives.

3.4 Splits

If an operation should be processed on several machines in parallel, the HYDRA shop floor scheduling module allows for the operation to be split into several "splits". These splits are handled by HYDRA like separate operations and can separately be logged on and off independently of each other. The collection of all actual data then pertains to the particular splits and to the original operation (master split), which is known to the PPS system.

3.5 Multiple machine production

If no planning tool is used in HYDRA (e.g. HYDRA shop floor scheduling module), the system can be configured in a way that allows for an operation to be processed, i.e. logged on, on several machines at the same time, without having to be split. Data is collected separately for every machine. A disadvantage of this variant is that planning becomes very difficult.

3.6 "Mixed operation"

As regards the shop floor, HYDRA generally supports all forms of mixed operation. Consequently, splits, merged operations and "normal" operations can be processed serially, in parallel or as multi-machine production. Data collection refers in each case to the collection forms presented above.

In this way, it might be possible in HYDRA, for example, that while an upper and lower part are produced, the operation called "compression lower part" can be logged on to another workstation at the same time for reworking purposes (e.g. debur).

4 Erfassungsarten der BDE (personenbezogen)

4.1 1.1 Summary

The sections that follow describe the different personnel-related views by way of examples. Postings based on people and on operations are displayed. To simplify matters, breaks are not included in the calculation of time intervals.

4.2 Single machine operation

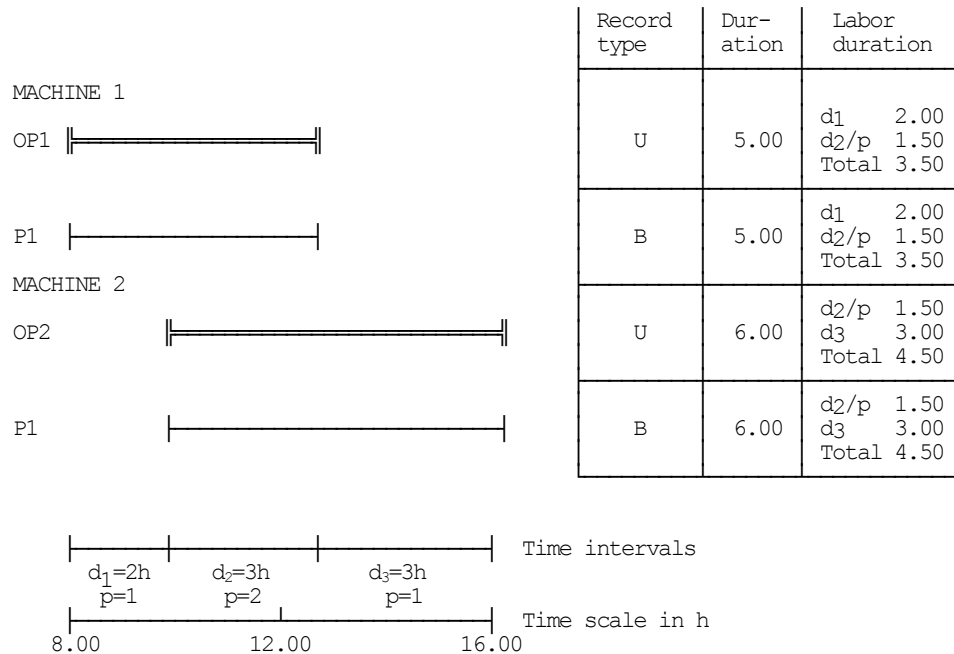
"Single machine operation" means that one person works alone on an operation at one workstation.

4.3 Multiple machine operation

Multi machine operation means that one person processes several operations at several workstations at the same time. This is normal, for example, on semi or fully automatic machines, where one person monitors several machines, feeding them with raw material and removing the finished parts. The labor utilization relating to operations is reduced according to the number of machines that are operated simultaneously.

Example

One person processes two operations at the same time on two machines.



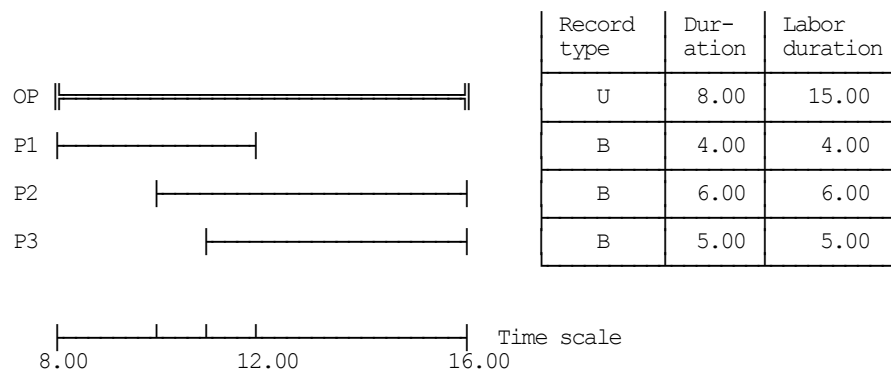
p : the number of times the same person has logged in
d_i: time interval i

4.4 Group work

Group work means that several people process a single operation at one workstation. With every additional person who is logged on, the labor utilization increases accordingly.

Example

Three people process an operation on one machine.



4.5 "Mixed operation"

HYDRA allows for all of the above-mentioned variants to be mixed, that is, HYDRA correctly handles the situation where a person with multi-machine operation is additionally logged on to a machine with group work.

Example

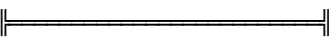
Several people process several operations on several machines (parallel production).

Machine 1 is an individual workplace

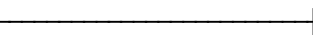
Machine 2 is a group workplace

Machine 1 Individual Workplace

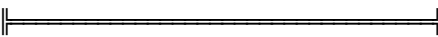
OP 1



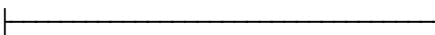
P1



OP2



P1



Record type	Dur- ation	Labor duration
U	d1 2	d1 2.0 P1
	d2 1	d2/2 0.5 P1
	d3 2	d3/3 0.7 P1
	Total 5	Total 3.2
B	d1 2	d1 2.0
	d2 1	d2/2 0.5
	d3 2	d3/3 0.7
	Total 5	Total 3.2
U	d2 1	d2/2 0.5 P1
	d3 2	d3/3 0.7 P1
	d4 1	d4/2 0.5 P1
	d5 2	d5/2 1.0 P1
	d6 1	d6 1.0 P1
	Total 7	Total 3.7
B	d2 1	d2/2 0.5
	d3 2	d3/3 0.7
	d4 1	d4/2 0.5
	d5 2	d5/2 1.0
	d6 1	d6 1.0
	Total 7	Total 3.7

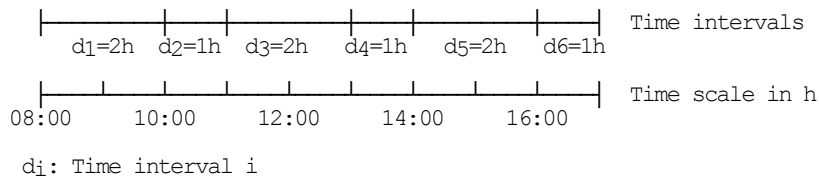
Machine 2 Group Workplace

OP 3

P2

OP 4

P1



U	d1	2	d1	2.0	P2
	d2	1	d2	1.0	P2
	d3	2	d3	2.0	P2
	d4	1	d4	1.0	P2
	Total	6	Total	6.0	
B	d1	2	d1	2.0	
	d2	1	d2	1.0	
	d3	2	d3	2.0	
	d4	1	d4	1.0	
	Total	6	Total	6.0	
U	d3	2	d3/3	0.7	P1
	d4	1	d4/2	0.5	P1
	d5	2	d5/2	1.0	P1
	Total	5	Total	2.2	
	Total	5	Total	2.2	

5 Erfassung in der BDE

5.1 1.1 Summary

In HYDRA data is generally recorded by collecting and processing dialogs. Processing of a dialog results in events that are the basis for updating (calculation of statuses) and posting (generation of log records).

Core functions of data acquisition are the functions to resolve dialogs into events as well as checking and processing functions that lead to the generation of events within the system.

Dialogs

Dialogs represent the system's interface with the shop floor.

Events are generated by resolving the dialog. Resolving of the dialog into individual events is performed based on dialog data, the processing logic of the dialog (incl. configurations) and the context of the dialog (posting status).

Some of the HYDRA-BDE input dialogs are:

Dialog	Meaning
A_AN	Log operation on
A_P_AN	Log operation and person on (together)
A_TR	Confirm/upload OP partially
A_UN	Interrupt operation
A_AB	Log operation off
P_AN	Log person on
P_AB	Log person off
P_AAB	Log all persons off (1... n P_AB)
SA_AN	Log merged operation on
SA_TR	Partial upload of merged operation
SA_AB SA_ABME	Log merged operation off

Events

Events result from resolving dialogs and are thus the result of data input. Which events get to the system at what times, as a result, represents an important point of controlling the software.

Further, partly optional, data acquisition functions are:

- logging of the dialog and the resulting event,
- provision of the posting result,
- provision of current information in the form of lists (e.g. machine lists, order lists),
- escalations.

6 Validation Checking

6.1 Summary

Validation checking of a dialog verifies whether the system is in the position to completely process all resulting events in the current status. If the dialog cannot be processed, i.e. not all events have been checked successfully, the dialog is rejected without making changes to the system.

These checks evaluate dialog data, the generated events and the HYDRA dataset and interpret the posting status. Furthermore, different (partly optional) configurations are taken into account.

Examples of **validation checks based on dialog data**:

- > if an OP is logged on, the registered OP has to exist within the HYDRA dataset
- > if an OP is logged on, the registered machine has to exist within the HYDRA dataset
- > if personnel is logged on, the person who performs the posting needs to exist in the HYDRA dataset
- > when posting to group workplaces, the person has always to be logged on as well

Examples of **validation checks regarding the posting status**:

- if an OP is uploaded partially, the registered OP has to be logged on to the registered machine
- > if staff is logged off the person who performs the posting needs to be logged on to the registered machine.

Some examples of **configuration** options taken into account by validation checking:

- if an OP is logged on, the person carrying out the posting must be allowed to log the OP on
- > if an OP is partially uploaded, the overdelivery quantity may not be exceeded

Some of the validation checks are explained in more detail in the sections that follow:

6.2 Validation checks regarding overdelivery or underdelivery

6.2.1 Summary

Quantities that are posted when an operation is partially uploaded, interrupted or logged off can be checked as regards overdelivery. In addition, they may also be checked as regards underdelivery when an operation is logged off.

By default, quantity checking in HYDRA has generally been designed for the manual collection of quantities. On the one hand, it is only performed if a quantity $\neq 0$ is included in the posting dialog and, on the other, it is performed every time a quantity $\neq 0$ is posted. The same applies when it comes to overdeliveries, even if they were confirmed beforehand by entering a deviation reason.

The overdelivery/underdelivery check, also referred to as target quantity check, can be activated in relation to operations as well as to staff.

6.2.2 Overdelivery/underdelivery checking activated in relation to operations

An overdelivery or underdelivery check can be enabled for the operation. The following values/settings are relevant in this context:

Underdelivery

Value in percent that may deviate from the target quantity. Example:

- Target quantity of the operation: 120 items
 - Underdelivery: 84%
- The actual quantity must not fall below 70 pieces.

Overdelivery

Value in percent that may deviate from the target quantity. Example:

- Target quantity of the operation: 120 items
 - Overdelivery: 168%
- The actual quantity must not exceed 140 pieces.

Reaction

In case the limit values of the overdelivery or underdelivery fields are exceeded, a possible response may be to display a warning or an error message. Possible values are:

"blank" no reaction

W warning

X error

Provided that a warning is activated ("W"), a quantity variance may be confirmed at the Windows terminal by entering a deviation reason. If an error is enabled ("X"), a quantity variance will be rejected.

Please note

By default, checking based on operations applies to the collected yield (primary quantity unit).

However, by customizing the processing code, it may also apply for scrap, rework or the problem quantity. Alternative quantity units (e.g. secondary quantity, etc.) are not planned to be checked by default.

Basically: DOS terminals generally reject any quantity variance with an error, even if the option is set to "W".

MOC (console) does not provide this function that is available at Windows terminals.

If the SAP interface "PP-PDC" is in use, limit values for overdelivery/underdelivery are transferred as absolute values. HYDRA converts them into percentage values at the interface.

6.2.3 Overdelivery/underdelivery check activated in relation to staff

In addition or as an alternative, it is also possible to enable a personal overdelivery/underdelivery check. For this purpose, the HR master provides the "target quantity check" option in the "BDE" tab. Possible values:

1) No check.

2) Logoff of order: A check is only performed on the logoff of an operation to verify whether the current yield is between the specified minimum and maximum target quantity, when the operation is logged off. Both quantities are also defined with respect to staff in the HR master.

3) Underdelivery/overdelivery: All quantity postings are checked for overdelivery, when orders are logged off/partially uploaded and interrupted. When an operation is logged off, it is also checked for underdelivery.

Please note

In general, personal checks only refer to yield (primary quantity unit).

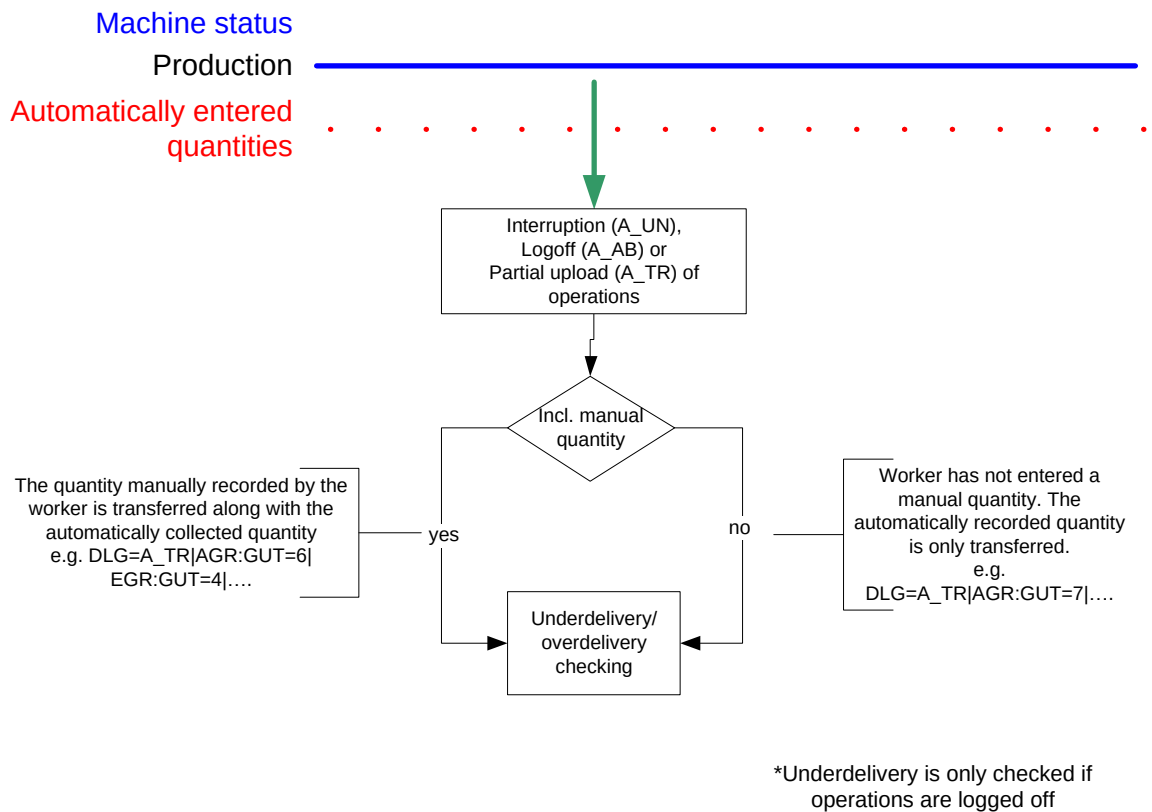
The limit values for overdelivery and underdelivery defined in operation data do not affect this and are checked separately - provided that they have been set accordingly. In this case, the personnel-related target quantity check is performed, then the operation-related check.

6.2.4 Overdelivery/underdelivery check for the automatic collection of quantities

In HYDRA standard, the target quantity check has generally been designed for the manual collection of quantities. It is only performed if a quantity $\neq 0$ is included in the posting dialog. The paragraphs that follow describe different scenarios that explain processing in connection with the automatic entry of quantities.

Scenario 1

If an operation is posted (interruption, logoff or partial confirmation/upload) in the active "production" status, the quantities that have been recorded automatically since the last posting (e.g. status change, operation posting, personal posting) and the quantities that might have been recorded manually are sent to the server for updating. As the command (DLG=...) sent to the server includes a quantity, a validation check is performed, irrespective of whether the worker has additionally posted a manual quantity $\neq 0$. The check now refers to the automatically collected quantity (AGR:GUT=...) as well as to the quantity that might be collected manually (EGR:GUT=...).



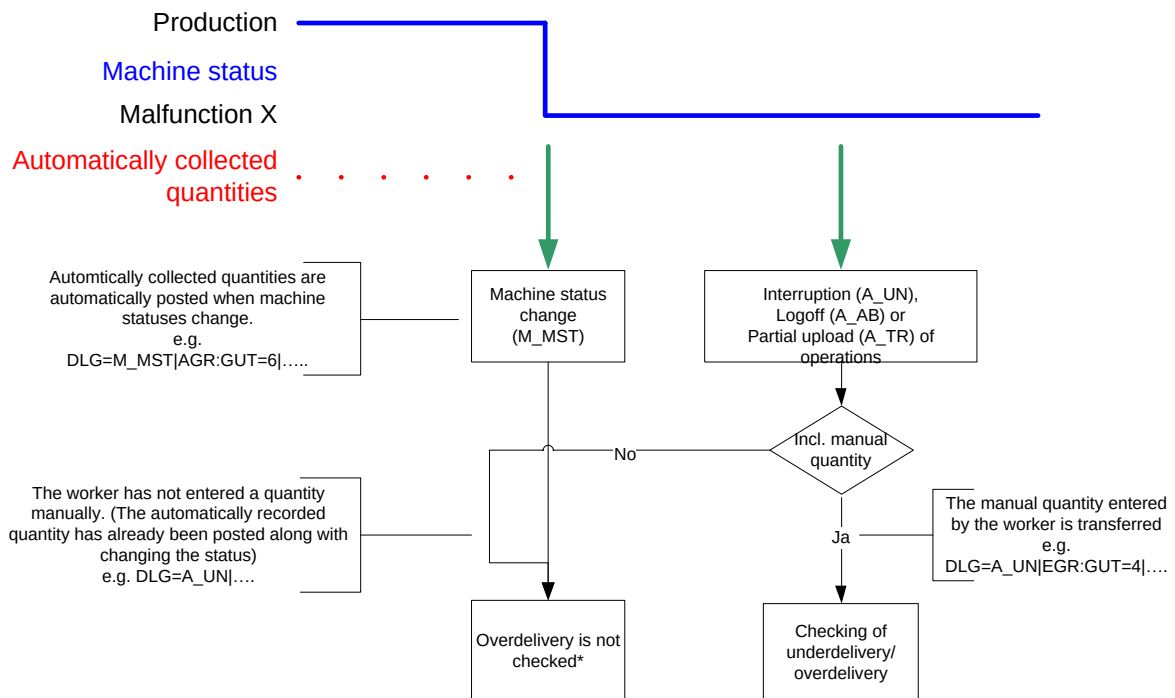
Scenario 2

If a machine switches from the production status to “malfunction” the quantities, which have been recorded up to this point in time (AGR:GUT=...) are sent along with the status change (DLG=M_MST|...) to the server for updating. As this is an automatic posting (status change), a validation check is not required.

In case an operation is posted afterwards, a check is only performed if the worker has manually recorded a quantity $\neq 0$. This quantity is sent to the server along with the posting (e.g. DLG=A_UN|EGR:GUT=...).

If the worker, for example, does not enter a quantity when the operation is interrupted (quantity = 0), no checking is performed, as no quantity is sent to the server (DLG=A_UN|...).

As automatic quantities have not been collected since the last status change, this posting does not include any automatic quantities, which could be checked for validity.

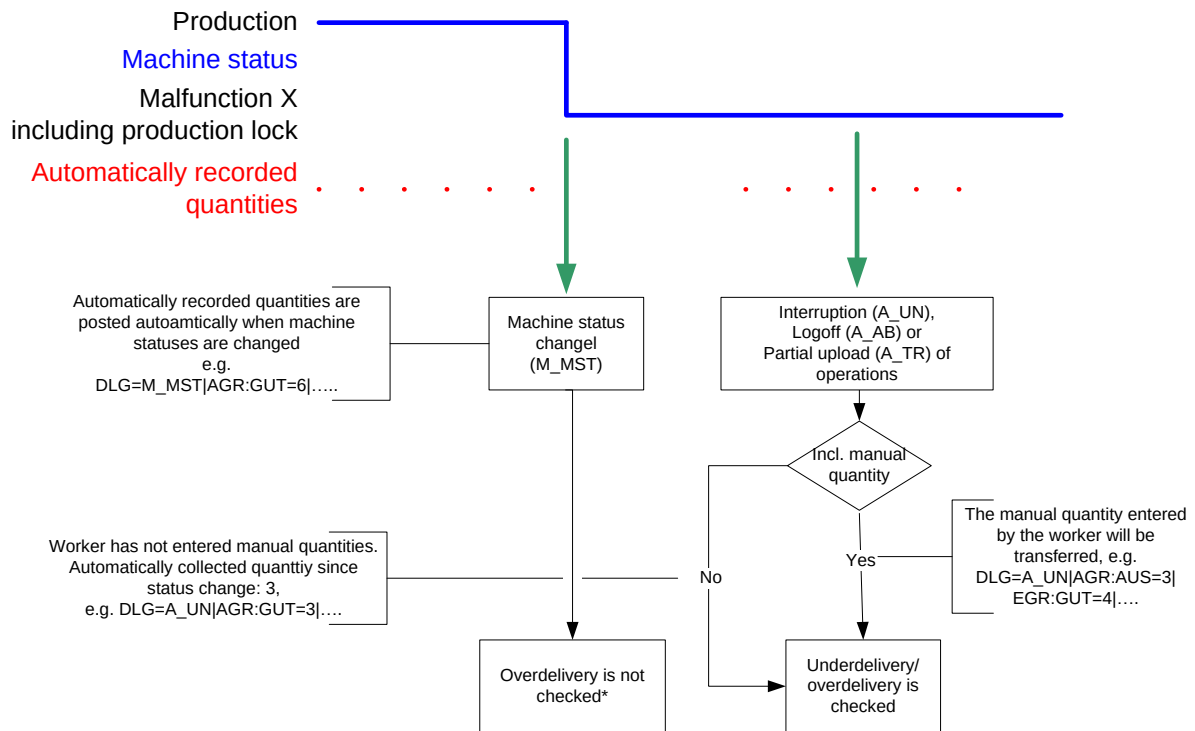


Scenario 3

Automatic quantities can still be collected, even if the machine is in the malfunction status and the production lock is set.

In this case, this automatic quantity is sent along with the manual posting to the server using the command (e.g. DLG=A_UN|AGR:GUT=...) and is checked for validity.

This is performed just as described in scenario 1, irrespective of whether the worker has entered quantities manually or not during the posting process.



7 BDE Log Records

7.1 General

The "posting" function generates log records from recorded events. Log records describe a period of time and include the evaluation of different values, such as quantities, durations or performances.

A BDE log record is an evaluated data record that is generated due to posting events. "Evaluated" means that this data record

- generally refers to a period of time (in an extreme case, this period of time is reduced to a point in time),

and

- includes durations (machine times, labor utilization, resource performance accounts) or
- evaluated quantities (e.g. yield, scrap)

HYDRA-BDE log records describe postings based on operations, staff and batches. Which log record is generated at which point in time depends on different factors, such as triggering posting events, configured posting rules or the HYDRA modules in use. The different types of log records are differentiated by their record type. The following record types based on orders are distinguished:

7.2 Record type A

General

A log record of the record type A is generated when an operation is logged on. It has been designed to be able to upload operation logons to the higher-level ERP/PPS system.

Triggering events and dialogs

Events: Logon of operation (A_AN)

Dialogs: Log operation on (A_AN),

Log operation and person on (together) (A_P_AN),

Log operation on with output batch (A_AN_MPL),

Beginning of shift (A_AAN)

etc.

Special remarks

Using optional configurations, different other dialogs can also cause an operation to be logged on (e.g. change order if the machine status changes).

The log record is not displayed in the maintenance of postings function. Consequently, it cannot be edited (changed, deleted).

7.3 Record type T

General

A log record of the record type T is generated when "partial uploads/confirmations" are recorded. In contrast to all other record types that refer to a period of time, this log record always refers to exactly one point in time (beginning of posting and end of posting are identical in the log record). Thus, this log record does not include durations, i.e. fields relating to durations do not include a value (0). The log record contains the quantities recorded with the upload.

Triggering events and dialogs

Events: all events that include a manual quantity (A_TR, A_UN, A_AB, etc.)

Dialogs: all dialogs that include a manual quantity .

Special remarks

In general, a partial upload may be performed for an operation that is logged on. However, the "posting onto OPs that are not logged on" configuration (workplace/machine configuration) allows for an inactive operation to be uploaded by way of a "quantity upload".

All quantities posted onto an operation are included in the single partial uploads (record type T) and as total in the log records "interruption of operation" (record type U) and "logoff of operation" (record type E). Thus, a T record is generated for each quantity that is collected and posted on an operation.

Manual quantities may be recorded in any dialogs, e.g. in a partial upload, when personnel is logged off, when an operation is interrupted or logged off, when batches are posted, etc. The log record of the record type T with manual quantities is directly generated, when the dialog is recorded/posted (not only when the operation is logged off/interrupted, for example).

In addition, the automatic collection of counters allows for quantities to be posted onto active operations. T records are generated for these quantities as well. HYDRA accumulates the quantities from the automatic collection over a period of time and generates the corresponding T record(s), **but only if the scenario changes**. This may be, for example, an order or person that is logged on manually, a manual, partial upload/confirmation or a manual or automatic machine status change.

The single partial uploads/confirmations also represent the collection of quantities with different reasons (e.g. scrap reasons).

Example:

Scrap (quantity 1) with the reasons 1, 2 and 3 each was recorded for an operation. The U or E record shows a total scrap quantity of 3. In addition, HYDRA records three T records each with a scrap quantity of 1 and the corresponding reason.

7.4 Record type U

General

A log record of the record type U is generated, when an operation is interrupted. Either the user can interrupt the OP manually or it can also be made automatically, for example, by the "shift automatic" of HYDRA-MDE at the end of the shift.

The log record of an operation includes the period of time between logon and interruption of the operation. The durations included in the log record refer to exactly this period of time and are synchronized with the shift model of the workplace to which the operation was logged on: the times are distributed onto the individual resource performance accounts according to the workplace/machine statuses that occurred during this period of time. The breaks that are included according to the shift model are posted onto the resource performance account 12.

The quantities and times posted in the log record are also posted onto the operation status.

Triggering events and dialogs

Events: Interruption of operation (A_UN),

Quantity upload to an inactive operation (A_MR)

Dialogs: Interruption of operation (A_UN),

Quantity upload to an inactive operation (A_MR)

End of shift (A_AUN)

etc.

Special remarks

In general, an operation may only be interrupted if it was logged on before. However, the "posting onto OPs that are not logged on" configuration (workplace/machine configuration) allows for an inactive operation to be uploaded using a "quantity upload". In this case, the log record refers to a point in time and, as a result, does not include any durations.

An operation may also be triggered to be interrupted even by different other dialogs through optional configurations (e.g. order change when the machine status is changed or automatic interruption of the operation when the target quantity is reached).

7.5 Record type E

General

A log record of the record type E is generated, when an operation is logged off. As regards content, this data record corresponds to a log record of the record type U.

Triggering events and dialogs

Events: Operation is logged off (A_AB),

An inactive operation is terminated (A_BE)

Dialogs: Operation is logged off (A_AB),

An inactive operation is terminated (A_BE)

etc.

Special remarks

If an inactive operation is finished the log record refers to a point in time and, as a result, does not include any durations.

An operation may also be triggered to be logged off even by different other dialogs through optional configurations (e.g. the operation is logged off automatically, when the target quantity is reached or predecessor operations are logged off automatically).

7.6 Record type H

General

A log record of the record type H is generated, when an (output) batch is changed/logged off (ADE-CHV, MPL). Either the user can change the batch manually or it can also be made automatically, for example, by a machine signal or by interrupting the operation or by the "shift automatic" of HYDRA-MDE at the end of the shift.

The log record includes the period of time between logon and logoff of the batch to/from the machine and operation. The durations included in the log record refer to exactly this period of time and are synchronized with the shift model of the workplace to which the batch and operation were logged on: the times are distributed onto the individual resource performance accounts according to the workplace/machine statuses that occurred during this period of time. The breaks that are included according to the shift model are posted onto the resource performance account 12.

The quantities posted in the log record are also posted onto the batch.

Triggering events and dialogs

Events: Logoff of (output) batch (CA AB)

Dialogs: (Output) batch change (CA_WL)
 Interruption or logoff of operation (A_UN, A_AB),
 End of shift (A_AUN) etc.

Special remarks

A batch posting does neither include labor utilization nor personal resource performance accounts.

A batch may also be triggered to be posted even by different other dialogs through optional configurations.

7.7 Record type B

General

A log record of the record type B is generated if a person is logged off from a workplace and an operation that is active at this workstation. Either the user can log the person off manually or it can also be made automatically, for example, by interrupting the operation or by the person clocking out (HYDRA-PZE: out) or by the "shift automatic" of HYDRA-MDE at the end of the shift.

The log record includes the period of time between the person being logged on and logged off to and from the workplace and operation. The durations included in the log record refer to exactly this period of time and, subject to the respective configuration, they are compared with the shift model of the workplace to which the person and operation were logged on or with the person's BDE shift model: the times are distributed onto the individual resource performance accounts according to the workplace/machine statuses that occurred during this period of time. The breaks that are included according to the shift model are posted onto the resource performance account 12.

Triggering events and dialogs

Events: Log person off (P_AB)

Dialogs: Person is logged off (P_AB)
 Interruption or logoff of operation (A_UN, A_AB),
 End of shift (A_AUN)
 HYDRA-PZE out (P_GEH)
 etc.

Special remarks

A person may also be triggered to be logged off even by different other dialogs through optional configurations (e.g. staff is logged off when the machine status is changed).

7.8 Special features

Evaluation date

The field "evaluation date" is defined for the work day evaluation of PZW (Personnel Time Management) and the wage calculation of the LLE module. It has been designed to assign postings to a settlement day for the LLE module and for the BDE/PZW comparison. Due to flexible working times, this settlement day may differ from the ADE shift date, in particular for night shifts.

Further information can be found in the document entitled [GLOSSARY_EvaluationDate.pdf](#).

Corresponding shift

The time stamp for logging off pertaining to a BDE log record determines the shift (shift date, shift number) to which the log record belongs.

Operation splits

When it comes to split operations, the log record is ONLY generated for the split operation. A log record is not generated for the split master itself. The status of the split master is only updated.

Merged operations

Log records are generated for merged operations as well as for individual operations. The statuses of individual operations are also updated accordingly. While posting, the recorded quantities and times of a merged operation are distributed to the corresponding individual operations according to different configurations. The log records of individual operations only include proportionate quantities and times.

8 Updating of the Posting Status

8.1 1.1 Summary

The functional area "updating of the posting status" has been designed to calculate statuses on the basis of recorded events. A status describes the condition of an object regarding the values that vary over time (e.g. OP status, posting status, etc.) and the relations between objects (e.g. OP is logged on to machine, person is logged on to OP and machine).

Usually, posting relations are generated for logons (operation to machine, person to machine and operation) and canceled when they are interrupted/logged off.

An object's posting status is updated every time when the event to be posted directly or indirectly affects the object.

Examples:

- By logging an operation on, this registered operation is set to the "running" status. The "operation logon" event directly affects the operation.
- By way of automatically recorded machine counters, resulting quantities are posted onto the operations that are currently active at the machine. The "automatic counter posting" event indirectly affects the operation(s).

The posting status also has an essential influence on which events are triggered by a dialog. Thus, the "log all persons off" dialog (P_AAB), for example, determines all persons who are active at the machine from the posting status and generates or posts a "personnel logoff" event for each person.

9 Posting of Times

9.1 General

The machine duration (or just duration) is determined by the time interval between logon and logoff of an operation. The machine scheduling time is harmonized with the shift calendar of the machine, whereas planned shift breaks are removed from the time interval calculation.

The labor utilization (or labor duration) is the sum of all labor times for each operation. This period is determined by the time interval between logging the user(s) on and off. The shift model of the workplace and the breaks included in this model are the basis even for personal postings (record type B). If several persons process several orders at the same time, the working time will be allocated proportionately for the corresponding operation in HYDRA according to the number of operations to which the persons are logged on.

The determined times (duration, labor utilization) are updated in the operation status (actual duration, actual labor utilization) and saved within the posting results - the log records.

Within the log record, assignment and meaning of time-related fields depend on the record type of the log record:

Field	Record type U/E	Record type H *)	Record type B
Duration	The operation's production time on the workplace	The output batch's production time on the workplace	The person's sheer registration time on the operation.
Labor utilization	Total of labor utilization of all persons logged on to the operation.	No labor utilization is defined in the log record for batch postings.	The person's labor utilization at the operation = evaluated registration time depending on the person being logged on (number of operations and workplaces).
RPA based on orders	The operation's production time on the workplace	The output batch's production time on the workplace	The person's sheer registration time on the

Field	Record type U/E	Record type H *)	Record type B
	distributed onto resource performance accounts. Please note: Subject to the current machine status, the resource performance account, which has been assigned to this status during the configuration, is "charged".	distributed onto resource performance accounts.	operation distributed onto resource performance accounts.
Personal RPA	Total of the personal resource performance accounts (see record type B) of all employees logged on to the operation.	Personal resource performance accounts are not defined in the log record for batch postings.	By way of the "proportionate RPA posting in personnel postings" option, the person's resource performance accounts (just as it is the case for the labor utilization) can be calculated proportionately - depending on the person's registration (number of operations and workplaces). If this option is not active personal RPAs correspond to RPAs based on orders.

*) Batch posting (only with operations that are subject to management in batches).

Time-related fields do not play a role with log records of the record type A and T (always 0) and therefore they are not included in this overview.

The document entitled "data collection in HYDRA-BDE (personal)" explains the process of determining times within personnel postings in different scenarios using examples.

Please note:

The determination of durations when status postings are recorded online at the terminal (or via PDM) only takes into account a limited number of shifts relating to the point in time of the last posting made for this machine.

However, durations might be missing or faulty in BDE log records if no postings (order postings, personal postings, status changes, shift changes, or similar) are entered for a machine over a longer period of time (e.g. the terminal is shut down).

9.2 Special features

Logging staff off during status 30000 "not assigned"

If a person logs off in status 30000 "not assigned", the duration of the labor utilization will be posted onto the resource performance account that is assigned to the "general disturbance" status within the personal BDE log record (record type "B"). If then the machine status is changed, the machine is assigned this new status over the complete period of time. Thus, resource performance accounts might vary.

Specific configurations

The following configuration options may influence the above-mentioned posting of times/durations:

Proportionate posting for parallel OPs

Configuration based on workplaces (tab configuration > general).

Project time recording

When it comes to project time recording, the times to be posted are recorded manually instead of being calculated by the system.

Post production time to main utilization time (MUT) during break

Cross-system configuration in the basic parameter settings of HYDRA.

RPA to calculate labor times

Cross-system configuration in the basic parameter settings of HYDRA.

Proportionate RPA posting in personnel postings

Cross-system configuration in the basic parameter settings of HYDRA.

Synchronize labor utilization with the person's HYDRA BDE shift model

Cross-system configuration in the basic parameter settings of HYDRA.

HYDRA-ADE waiting period processing

Cross-system configuration in the basic parameter settings of HYDRA.

Merged operations

If several operations are grouped into a merged operation, the times are distributed onto the individual operations according to different configurations.

Split operations

Further information on how to post split operations or their split master can be found [here](#).

10 Posting of Quantities

10.1 General

Manually collected quantities

Manual quantities are always posted onto the registered operation and, subject to the respective configurations in HYDRA, to the person carrying out the posting as well. The below table shows whether or when manually collected quantities are posted onto the operation and/or the reporting person.

Posting	Posting to operation	Posting to reporting person
Interrupt OP	Yes	No *) **)
Log OP off	Yes	No *) **)
Partial confirmation/upload	Yes	Yes **)
Log person off	Yes	Yes

*) compare the "Notes on personal posting of quantities"

**) The person who performs the posting needs to enter his/her personal badge number in the posting dialog.

Please note for entering the quantity when personnel is logged off:

If several operations are logged on simultaneously, the quantities will be posted onto all operations.

Recommendation:

The relevant quantities based on operations should be recorded as partial upload for each operation, before the person logs off (without entering a quantity).

Automatically collected quantities

All OPs and employees logged on to the machine are assigned automatic quantities. Automatic quantities result, for example from counter collection.

The quantities to be posted are computed according to the partitioning/cavity that is respectively defined for the operation and operations or staff are posted.

Notes on the personal posting of quantities

The personal posting of quantities depends on the workplace on which an operation was produced. In general: automatically collected quantities are always posted onto all persons who are logged on.

Individual workplace (EAP)

If quantities are neither recorded automatically nor with respect to staff (i.e. no quantity is entered when a person logs off), no quantities will be taken over to the person. Consequently, the quantity fields within the personal posting dialog (posting of record type "B") are empty (0).

Exceptions:

- a) The *"quantity posting to staff"* configuration option within the machine/workplace configuration allows for the manually recorded quantity to be posted onto the person who has so far been logged on for the longest time, when an OP is interrupted or logged off.
- b) Using the option of the HYDRA basic parameter settings "post manual quantities as automatic quantities" allows for manually recorded quantities to be posted like automatically collected quantities, i.e. they are posted onto all OPs and persons logged on.

Group workplace (GWP/GAP)

At group workplaces, the quantity is always taken over to the person when the OP is interrupted or finished, as a unique assignment between person and operation has been established.

10.2 Special features

The following configuration options may influence the above-mentioned quantity posting:

Post manual quantities as automatic quantities

Cross-system configuration in the basic parameter settings of HYDRA.

Quantity posting to staff

Configuration based on workplaces.

Merged operations

If several operations are grouped into a merged operation, quantities are distributed onto the individual operations according to different configurations.

Split operations

Further information on how to post split operations or their split master can be found [here](#).

11 Controlling the Sequencing List

General

The sequencing list of the shop floor terminal is flexible and can be adjusted to meet specific requirements. The settings described below control how data is provided in the sequencing list. Please keep in mind that certain settings can only be made during the customizing process.

Workplace/ machine configuration: "Sequencing list" setting

Description

Which order pool is used to generate the sequencing list can be defined in the *sequencing list* configuration of the [workplace configuration](#) dialog.

The pool of orders is determined by planning in HYDRA shop floor scheduling or by the specifications from the PPS system and is a result of whether the OP was planned directly for a machine or a machine group.

Possible settings

S - Basic setting

The value is transferred from the option with the same name in the [HYDRA basic parameter settings](#).

M - Pool of workplaces

Only the operations planned for the workplace are displayed in the sequencing list at the terminal.

G - Pool of workplaces and groups

The sequencing list at the terminal displays those OPs that are either planned for the current workplace or for a different workplace in the group or that are still in the pool of groups.

K - Pool of workplaces and categories

The sequencing list at the terminal displays only the operations that are planned for workplaces in the same machine category.

H - Group control

The sequencing list at the terminal displays those OPs that are either planned for the current workplace or for a different workplace in the group.

Workplace/ machine configuration: "Number of OPs" setting

Description

The maximum number of OPs for the sequencing list can be configured by the option "number of OPs" within the [workplace configuration](#). The OPs are selected in ascending order based on how the sequencing list is sorted.

Possible settings

0 = no restriction (display all existing OPs)

1-999 = maximum number of OPs

Order types: "Sequencing list" setting

Description

The configuration is made at order level in the *sequencing list* option under [order types](#).

This setting can only be changed during customizing.

Possible settings

- J The OP should be displayed in the sequencing list.
- F The OP should only be displayed in the sequencing list if it is fixed.
- N The OP should not be displayed in the sequencing list.

Processing code: "Sequencing list" setting

Description

Whether or not an OP with this processing code should be displayed in the sequencing list can also be defined in the [processing codes](#) `..\..\functions\moc\MOC_ProcessingCodes.pdf` configuration.

This setting can only be changed during customizing.

Possible settings

- J The OP should be displayed in the sequencing list.
- N The OP should not be displayed in the sequencing list.

Order status assignment: "Sequencing list" setting

Description

Whether an OP should appear in the sequencing list based on its status can be defined using this option under [order status assignment](#). If an "N" is defined here, an OP in this status is not displayed in the sequencing list.

This is used as a standard feature to ensure that only prepared or interrupted OPs appear in the sequencing list.

Furthermore, it can be configured that running OPs also appear in the sequencing list.

This setting can only be changed during customizing.

Possible settings

- J The OP should be displayed in the sequencing list.
- N The OP should not be displayed in the sequencing list.

Miscellaneous notes

Generally, the following operations never appear in the sequencing list:

- Locked operations
- Individual operations from merged operations that were generated at MOC.
- The original operation for split operations
- Operations of inactive alternative sequences

Order of operations in the sequencing list

Generally, the data listed below defines how the operations are ordered in the sequencing list at the terminal:

1. Internal field for the planned start date (auftrags_bestand.sort_dat)
2. Internal field for the planned start time (auftrags_bestand.sort_dat)
3. HYDRA combined order/ operation number (auftrags_bestand.auftrag_nr)

The system fills in the internal fields for the planned start date and planned start time. This depends on how the operation was created in the system (transferred via the interface or manually created in the system) and on the planning functions used subsequently. How the system does this is described in each applicable documentation.

- [Manual editing via the user interface or via the system interface – initial data creation](#)
- [Manual editing via the user interface or via the system interface - modification](#)
- [Using HYDRA Shop Floor Scheduling](#)
- [Using HYDRA Order Sequencing](#)



When using order sequencing, we recommend to only use the "M" option (pool of workplaces) to configure the sequencing list; the order within the sequencing list at the terminal is not defined when the other options are used.

12 Order and Operation-Related Functions

Usage

The functions described here allow the performance of order and/or operation-related functions.

Integration

Functions are called in various applications. These applications are listed in detail in the descriptions of various functions.

Prerequisite

Orders/operations must be defined in the system. The prerequisite for using the individual functions are listed in the function descriptions.

Change Order Status

Function authorization	or.statchg	Change Order Status
------------------------	------------	---------------------

This function may be used to convert the status of a complete order. The function is integrated in the following application:

- Order overview → Order tab → Category 'Change status'

The dialog contains the following fields

Order

Order number

Current status

Current status of the order

Action

The action available depends on the current status:

Current status	Possible new status
Not free (specific status only available under customizing)	- Release order (default) - Set order to "in process" - Terminate order
Prepared	- Set order to "in process" (default)

Current status	Possible new status
	- Terminate order
Started	- Terminate order

Setting the status to 'Terminated' is only permissible if no operation is running. As soon as an order status is set terminated, the status of all other operations of this order is also set terminated.



If an order is terminated by using this function, a log (record type E) is only generated for the associated operations, if the option "Generation of log record for terminate OP (MOC)" is set. The workplace used for generating the log is that registered for the operation (regardless of whether or not the operation has been scheduled for this workplace). If no workplace has been registered, the workplace with the lowest number within the group is identified.

Update order

Function authorization	or.actualize	Update order
------------------------	--------------	--------------

The function "Update order" is used to update one or several marked orders. The action for updating an order is performed directly; there is no additional window opening so that the action is confirmed.

After updating, the order data will be updated so that potentially updated data will be displayed in their updated condition.

The function is integrated in the following applications:

- Order overview → Order tab → Category 'Execute'
- Order information → General tab → Category 'Execute'
- Process operations → Other functions → Category 'Change status'

Terminate order

Function authorization	or.terminate	Terminate order
------------------------	--------------	-----------------

The function "Terminate order" is used to terminate one or several selected orders. The action for terminating an order is performed directly; there is no additional window opening so that the action is confirmed.

It is also possible to call the termination for one or several selected operations. In this case, the orders of these operations are terminated.

After updating, the order data will be updated so that potentially updated data will be displayed in their updated condition.

The function is integrated in the following applications:

- Order overview → Order tab → Category 'Execute'
- Order information → General tab → Category 'Execute'

Change operation status

Function authorization	op.statchg	Change operation status
------------------------	------------	-------------------------

This function may be used to convert the status of an operation. Whether or not the status of an operation can/may be changed also depends on its current (previous) status.

Operations in the 'Running' status may not be changed through the "Change status" function, but may only be changed by "Interrupt OP" or "Log OP off". An operation which was interrupted may only be reset to 'Running' by using the function "Log OP on".

The dialog contains the following fields

Order / Sequence / OP / Split

Operation number

Current status

Current status of the operation

Action

The action available depends on the current status:

Current status	Possible new status
Not free (specific status only available under customizing)	- Released (default) - Terminate
Prepared	- Terminate
Interrupted	- Terminate

Depending on the option "Generation of log record for terminate OP (MOC)" in the HYDRA basic settings, in addition to setting the operation to status "E", an appropriate end posting (log of record type "E") is generated when an operation is terminated and may also be reported to the higher level ERP/PPS system. This takes place for all operations when an order is terminated.

The workplace used for generating the log is that registered for the operation (regardless of whether or not the operation has been scheduled for this workplace). If no workplace has been registered, the workplace with the lowest number within the group is identified.

This function may also be used to terminate overhead cost operations/orders which may not be terminated by using the "normal" dialogs A_AB at the terminal and MOC.

The status of a waiting period operation may be changed by "Change status". However, the status change is not important, because a waiting period operation is not an operation in which an activity is carried out. The status changes are a mere information for the user. The reporting functions on the screens are not relevant for waiting period operations. This also applies to error messages that might appear when logging on or off.

Change secondary status

Function authorization	op.secstatchg	Change secondary status
------------------------	---------------	-------------------------

The button "Change secondary status" may be used to change the secondary status of an operation. The function is only available if the current status of an operation has been configured with inclusion of the option "Editable secondary status" (Customizing).

The function is integrated in the following application:

- Order overview → Operation → Category 'Change status'

The new secondary status may be entered manually or selected from a shortlist. Please note that the field is compulsory; deleting the current status is not possible. If a secondary status is to be "reset" after having been set, an appropriate secondary status is to be provided for this.

Change resource status

Function authorization	op.resstatchg	Change resource status
------------------------	---------------	------------------------

The function "Change resource status" may be used to change the resource of an operation. The function is integrated in the following application :

- Order overview → Operation tab → Category 'Change status'

Please note that all resource statuses are always set simultaneously.

Lock operation

Function authorization	op.lock	Lock operation
------------------------	---------	----------------

The function "Lock" is used to lock one or several selected operations. A locked operation may no longer be logged on in the BDE module (shop floor data collection) and will no longer appear in the terminal sequencing list, either.



Locking an operation will not have any effect on planning in the HYDRA shop floor scheduling module (HLS).

The function is integrated in the following applications:

- Order overview → Operation tab → Category 'Lock'
- Process operations → Other functions → Category 'Lock'

The action is performed directly, i.e. there is no additional window opening so that the action is confirmed. If a minimum of one operation is already locked among the selected operations, locking is implemented nevertheless. No error message is generated for the already locked operation(s). An error message is only indicated if an operation previously unlocked could not be locked.

Unlock operation

Function authorization	op.unlock	Unlock operation
------------------------	-----------	------------------

The function "Unlock" may be used to unlock one or several operations; after that, logging on the operation as well as displaying it in the sequencing list is possible again.

The function is integrated in the following applications:

- Order overview → Operation tab → Category 'Lock'
- Process operations → Other functions → Category 'Lock'

The action is performed directly, i.e. there is no additional window opening so that the action is confirmed. If a minimum of one operation is not locked among the selected operations, unlocking is implemented nevertheless. No error message is generated for the unlocked operation(s).

An error message is only indicated if the lock on an operation previously locked could not be canceled.

Please note: Operations that were split (so-called "Split masters") are also locked. These operations must not be unlocked through this function!

Reactivate operation

Function authorization	op.reactivate	Reactivate operation
------------------------	---------------	----------------------

The function 'Reactivate' may be used to reactivate terminated operations within the system. The function is integrated in the following application:

- Order overview

The function may only be operated for precisely one operation at a time. If several operations are selected, the function is not active. In addition, the function may only be operated for terminated operations.

After operating the function, the user has to confirm the reactivation in a dialog. By confirming, the terminated operation is reactivated: the operation changes into the status with control indicator *interrupted*. The operation may now be logged on again.

13 Day Types

Summary

Menu	Master data → Production control → Day types
Transaction code	dtmf
Function authorization	mddtmf

Day types must first be defined before the shift plan can be created for each work day. The planned working times and break times for up to four shifts for each day are defined in a day type.

Usage

The following points must be considered in advance before setting the definitions and configurations:



For days that have identical shift cycles, only one day model needs to be defined.



On the other hand, however, if work schedules are planned differently on different days (e.g. "short Friday"), then several (different) day models will be required.



Machines assigned to the same shift cycle, but with different breaks, require different day models.



Machines assigned to the same shift cycle, but with a different number of shifts (two shift operation, three shift operation) require different day models.



Only the last shift of a day model may include a day change (e.g. 10.00 pm - 6.00 am)!

If a day model is modified, this change will have an immediate effect on all machines that are assigned to this day model within the year model. Therefore, we strongly advise against making any changes to the day models currently in use (i.e. valid for the shift day on which the changes are made).



If, contrary to expectations, it should become necessary to make changes, then these changes may only relate to subsequent shifts (no longer affecting the current shifts). Furthermore, these changes must be made at least two hours before a shift change. Reason: different HYDRA components read the shift models in cyclical intervals and compare them to the shifts stored locally.

Requirement

The days off should be set up first.

Selection criteria

Day type ... to ...

Narrows down to day type number

Designation

Designation of the day type

Field descriptions

Day type

Unique day type identification

Designation

Designation of the day type

Shift time

Details about the beginning and end times for each shift

Please note: For technical reasons, a shift may not start or end at 12:00 am (midnight).

Type

At this time, this field is not relevant for processing.

Breaks

Details about breaks (beginning and end times) within the current shift.

When defining breaks, the times accrued in these periods are "cut out" from the RPA time accounts.

Number of personnel capacity

Production: This setting is only relevant in conjunction with planning group workplaces in the graphic planning board. In HYDRA shop floor scheduling, group utilization is calculated based on the defined shift calendar and the planned production capacities. If no personnel capacity is defined, the system implicitly assumes capacity 1.

Set up: This setting is currently not considered.

Autom. status change to status 999 (only relevant for MDE configuration)

A machine status can be assigned automatically using this indicator at the end of certain shifts (e.g. on Friday evenings) and vice versa this status can then again be cleared at the beginning of another shift (e.g. Monday mornings).

No activation: Automatic shift is not activated.

Activate after end of shift: At the end of a shift, the machine is automatically assigned to status 999. If no status change is made when the next shift begins, the status remains at 999 for the entire shift.

Please note: If at the end of a shift the status is set to "Production", this status remains set in the subsequent shift; in this case, status 999 is not assigned.

Deactivate at beginning of shift: At the start of a shift, depending on how the extended weekend automatic is activated for the machine configuration in the MDE configuration, the terminal switches automatically to either status "not assigned" or to the status that was defined prior to the status 999 activation. If status 999 is set, it is terminated.

Recommendation: To ensure that the MDE evaluations are not distorted, status 999, which, for example, signals a weekend, should be assigned to the resource performance account "Free breaks".

14 Days Off

Summary

Menu	Master data → Production control → Days off
Transaction code	wfmf
Function authorization	mdwfmf

The dates for all of the days off for a relevant year are entered in the days off list. The days off list is used to differentiate between workdays and days off when assigning day types to a year model.

Usage

Once the days off list has been created, it is available for all year models. This means you only have to create the list once at the beginning of the year and the data will be considered in all year models for the relevant year.

The days off list must be created before setting up the year models. If any entries are changed in the days off list at a later time, these changes can only be considered in those year models that are created after this change. Any year models that have already been created remain unchanged and may need to be adjusted by the user himself/herself.

Selection criteria

The application provides the following selection criteria:

Date from to

Limited by date

Field descriptions

Date

Date of the day off.

Description

Description

Number

Unique number for the day off

Year

Year for which the day off was defined

15 Year Model

Summary

Menu	Master data → Production Control → Year Model
Transaction code	ymmf
Function authorization	mdymmf

The year models serve as the machine's shift calendar.

Usage

The year model is required in order to assign a shift calendar to a machine/ workplace. To this end, each work day is assigned exactly to a day type in the year model. The days off defined earlier do not need to be considered (no shifts). Each machine can be assigned to exactly one year model.

The following points must be considered when doing so

- If a year model is modified, this change will affect all machines *indirectly* that are assigned to this day model within the year model.
- Any changes to the year model should only be carried out for days/ periods set in the future.
We strongly advise against making any changes to the year model for the current day.

A shift model is defined for each day in the year model, which HYDRA uses as the basis for calculating working time. Each workday is assigned exactly to one day type.

Requirement

The following master data must have been created before setting up a year model.

- The days off should be set up.
- The day types should be set up.

Selection criteria

The application provides the following selection criteria:

Year from to

Narrows down the selection to the year models for the chosen years

Field descriptions

Year model, designation

The current year model to be edited and its designation are shown at the top.

Factory calendar

The factory calendar identifier can be set for (a maximum of) one year model. Because it is used for planning/ [scheduling](#), a valid factory calendar must be defined for each year when using HYDRA shop floor scheduling.

The year model for which this identifier is set is of no further importance for data entry.

Graphic display

The day model assigned to each day in the separate fields of the year model are shown in the graphic display. The background colors used depict the following:

Monday – Friday	gray
Saturday, Sunday	orange
Days off	red
Current day ("today")	dark red

For each day, the assigned day type (day type number) is displayed. No day type is assigned to days on which no entries were made. HYDRA treats these days as "no shift" days.

Assignment

In the lower range are functions that can be used to assign day types to the year model. When doing so, you have the option of either assigning the day types by *weekday* or by *rhythm*.

The procedure used to assign a day type to a year model is described in the following chapter *Defining a year model*.

Editing functions

Defining a year model

Do the following to define a year model:

Select the editing function Insert; an editing dialog will open. Use the index tab *Weekdays* or *Rhythm* to now assign day types to the year model.

Weekdays

From ... to ...

Defining the period for which a day type should be assigned.



When making your selection in graphic display, all weekdays (Monday - Friday) are selected automatically that are in the above mentioned period.



Alternately, you can also select individual workdays. The selection then made in the graphic display is based on what you previously selected, in each case for the period mentioned above.



When making your selection in graphic display, all Saturdays and Sundays are selected automatically that are in the above mentioned period.



Alternately, you can also select the particular Saturday or Sunday. The selection then made in the graphic display is based on what you previously selected, in each case for the period mentioned above.

Include public holidays

If this is selected, all public holidays (defined as days off) are included in the selected period.

Example:

You selected *Weekdays* and *Include public holidays*. When a day type is assigned in the next step, all public holidays that fall on a weekday (Monday - Friday) will also be assigned to this day type.

Exclude public holidays

If this is selected, all public holidays (defined as days off) are excluded in the selected period.

Example:

You selected *Weekdays* and *Exclude public holidays*. When a day type is assigned in the next step, all public holidays that fall on a weekday (Monday - Friday) will not be assigned to this day type.

Public holidays only

If this is selected, *only* public holidays (defined as days off) are considered in the selected period.

Example:

You selected *Weekdays* and *Public holidays only*. When a day type is assigned in the next step, only public holidays that fall on a weekday (Monday - Friday) will be assigned to this day type. All other public holidays that fall on a weekend as well as normal weekdays are not assigned to a day type.

Day type

Enter the day type here that you would like to assign to the days you selected previously.



Click on this icon to assign the day type entered to the selected days. The day type will then appear in the graphic display for each selected day.



To delete a day type assignment, click on this icon.

Please note:

Alternately to entering a period and to defining the selected weekdays, you can also make a selection in the graphic display directly by hovering the mouse cursor over the first day, by then clicking and holding down the left mouse button and by now dragging the cursor to the last day where you then release the mouse button. All days that chronologically fall between the first and the last day are now highlighted.

Another option is to press the Ctrl key and clicking with the mouse on each day that you would like to include in the selection.

Rhythm

In this index tab you can define a rhythm that you would like to assign to the year model in a period (from - to or the days selected in the graphic display).

To do so, enter one by one the definition of the rhythm in which you would like to define the day type and the number of days (for example: Day type 4 for one day, day type 1 for three days, day type 2 for one day, day type 0 (no day type) for two days).

Then, accept the defined rhythm in the year model.

Type, duration

Day type and duration that should be transferred to the rhythm table.



Accept the type and duration in the rhythm table.



Delete the selected entry from the rhythm table.

From - to

Period in which the defined rhythm should be assigned.

Click on the icon to assign the days/rhythm previously selected to the selected days. The relevant day type will then appear in the graphic display for each selected day.

Method recommended for changes in the shift calendar

It may be necessary for reasons within the operation to make changes to the shift calendars. This might be the case, for example, when:

- The company switches its shift operation (e.g. from a two shift to a three-shift operation)
- Shift times are changed

Generally, the HYDRA shift calendar is a function that if configured or handled improperly, as is the case for all HYDRA configurations, can have unpredictable effects.

We strongly advise against making any changes to the day models currently in use (i.e. that are valid for the day the changes are made).

In this case, we recommend to follow the following procedure:

- As a rule, you should *not* wait until the day to make changes on which the changes will be active.
- Call up the dialog Day type.
- Create a *new* day type or change an existing, but not an active (i.e. not an assigned), day type. (Contrary to the above, we strongly advise against making any change to existing day types that have already been assigned to one or more year models.)
- Call up the dialog Year model for which the change is to be made.
- Assign the new day type. (When doing so, it is critical to keep in mind that the time as of which the new day type should take effect for the first time should definitely be in the *future*.)

16 Order Types

Summary

Menu	Master data → Order → Order types
Transaction code	ot
Function authorization	mdot

Usage

You use this function to create or modify order types in the system.

Integration

Order types are issued to structure the orders in accordance with their use. Each order type includes various control information that is decisive when managing orders. The possible order types are configured in HYDRA and as such, the settings define how they behave/perform in the system. The order type is set for the order header and also included in the operations (for information purposes).

Each order type is assigned to a category. This category combines similar order types.



We strongly recommend that order types only be copied within one category.

Selection criteria

The below-mentioned criteria can be used for selection within the selection panel:

Category

See field description

Active

This is a tri-state checkbox which enables to select either only active order types, inactive order types or both.

Field descriptions

General index tab

Order type

Definition of the order type. The parameters of this configuration control, among other things, performance within the system.

Designation

Designation of the order type

Active

If this indicator is set, then this order type is available and can be used. The indicator is checked when an order is first created or modified.

Category

The classification establishes a logical umbrella term covering similar order types. Possible values:

FA	Production order Standard production orders created e.g. by an interface in HYDRA are to be assigned to the "production order" category
PJ	Project order Projects are, among others, characterized by their uniqueness. The project steps that are to be performed as a part of a project can be combined in a project order
PM	Maintenance order Maintenance orders have been designed for the planning or collection of maintenance measures
KP	Capacity order Orders of this order type/category have been designed for capacity reservations in the HYDRA shop floor scheduling module (HLS). Consequently, data, for example, cannot be recorded for them.
GK	Overhead cost order Orders for the collection of overhead costs.

Symbol

Graphic that can optionally be assigned to the order type. It is displayed in several functions/reports (e.g. order overview).



The size of the graphic must not exceed 16x16 pixels and it has to be a file in the BMP format (16 colors).

Comment

Designation or description of the order type

Dialog control

Here you can enter a dialog control that deviates from how the system typically behaves and it can be referenced accordingly in the dynamic dialog. This makes it possible to show or hide situation-specific input fields at the terminal.

**Please note:**

This setting can also be defined at the workplace (machine/ workplace configuration). When a terminal dialog is opened, dialog control for both order type and workplace are considered. For this reason, we recommend that dialog control be defined either in the order type or at the workplace.

Editing inventory data

Reserved; currently not available for editing

Editing recorded data

Reserved; currently not available for editing

Planning index tab**Planning**

This setting determines whether and/ or how orders of this type are relevant for detailed scheduling. You can choose from following settings:

N No planning

T Scheduling only

The order is scheduled, that is the lead time is calculated based on the process times and the dates are set accordingly. No competitive situations are considered during (lead time) scheduling.

F Scheduling and detailed planning

The order is relevant for detailed planning and is consequently - in addition to scheduling - considered during automatic and/ or interactive planning.

Please note: During automatic dispatch (detailed scheduling), operations are dispatched to a workplace. In the process, the situation pertaining to capacity and assignment is considered for the workplace, which may result in the inability to dispatch by the planned date determined from lead time scheduling.

**Please note:**

Whether or which operations are in fact considered during scheduling or detailed planning also depends on the configuration of the same name in the *processing code* or the *order-related status configuration*.

Planning dates

How planning dates behave during (re)scheduling/re-planning

O Planning dates are not overwritten (default)

T Planning dates are overwritten by scheduling

Setup time in shop floor scheduling

Subject to the settings configured for the order type, the below-mentioned options are available to present the static setup time of interrupted operations in the graphic planning board (interactive planning):

- " " Target setup time
- R Remaining setup time: Target setup time – RPA07 (posted setup times)
- B In case the operation has already been started (i.e. it is now interrupted), a setup time of 0 is assumed.

Running operations do not take into account the setup time.

These settings only apply for interactive planning.

User field for requirements calculation formula

This field is only relevant in connection with the license for multiple assignments (HLS-MFB).

Further information can be found in the relevant [document](#).

Scheduling without implicit order release

This setting prevents order release (setting the status using the control indicator "V") during the first scheduling run.

Consideration of production variants when transferring OP

This processing is only active in conjunction with an HLS license for production variants.

- E The system determines whether a valid production variant exists at the time the OP is transferred/ created.
- P At the time an OP is transferred/ created, the data transferred is checked only to verify that it matches a valid production variant (validation check).
- N No determination/ validation check

Considering production variants in planning

This processing is only active in conjunction with an HLS license for production variants.

- E Determination. A valid production variant is determined at the time an operation is planned in graphic planning.
- P Testing only. At the time of planning, the data transferred is checked only to verify that it matches a valid production variant (validation check). Default values are not taken over from the production variant.
- G Group-specific testing. At the time of planning, the data transferred is checked to verify that it matches a valid production variant (validation check), as it is also the case with option "P". However, in contrast to the option "P", checking is only performed within the group defined for the operation. Default values are

not taken over from the production variant.

N Not used.

This processing can be controlled in more detail on the level of capacity groups. Further information on this can be found in the relevant documentation dealing with the configuration of [groups](#).

Processing index tab

Sequencing list

This setting determines whether operations for orders of this order type are displayed in the sequencing list at the terminal.

The following settings are available:

- F Operations that were fixed are displayed. The fixing indicator can be set manually in HYDRA shop floor scheduling.
- N No; orders of this order type are generally not displayed in the sequencing list at the terminal. This makes sense, for example, for overhead cost operations.



Please note

With the exception of the option "N", for all other options, in addition, the configurations "sequencing list" with the same name are always taken into account in the *processing code* or in the *order-related status configuration*.

Locked OPs are generally not displayed.

Recordable

This identifier defines whether operations for orders of this order type can be recorded, i.e. posted.

- J Yes, operations can be recorded.

Whether an operation is really recordable also depends on the setting with the same name in the processing code.
- N No; Operations may not be recorded. An attempt to log any such operation on is rejected with a validation error.

Combined logon of order

This identifier is used for displaying "overhead cost operations".

- J With this order type, operation and person are firmly linked with one another. Its behavior (posting) is the same as with a posting to a group workplace (GAP).
- T No overhead cost processing.

Logging OP on again at shift start

This identifier defines operation logon behavior at the start of a shift.

- | | |
|---|---|
| J | Automatically log operation back on at start of shift. |
| N | Do not log an operation back on at start of shift. |
| X | The processing depends on the appropriate setting in the workplace/ machine configuration |



Please note:

This processing is only relevant for "MDE workplaces". An MDE workplace must meet the following conditions:

- The workplace must be assigned to a terminal
- The terminal must be configured in MDE operation mode

May OPs of the order be finished?

This identifier defines whether OPs of this order type may be finished. The following settings are available:

- | | |
|---|--|
| J | Yes, OPs may be finished. |
| S | Checking status of predecessor. In this case, the status of the preceding OP must have the identifier "Successor can be logged off" set. |
| N | No, OPs may not be finished. A validation check is done when an OP is logged off. |



Please note:

It can be useful for overhead cost orders to configure them so that they cannot be logged off using the posting dialog *Log OP off*, since their operations should exist as cost collectors over a longer period of time. In this case, this identifier can be set to "N" so that an OP logoff is prevented.

Log person off when shift ends

This identifier is used for detailed configuration of personal data entry at MDE workplaces. Since fully automatic shift ends are generated via the terminals when HYDRA MDE is used, here the user can set whether the persons logged on to the workplace should be automatically logged off or remain logged on at the end of a shift.

- | | |
|---|--|
| J | Always log staff off when the shift ends |
| N | Always save staff when the shift ends except for manual logoff |

- X The processing depends on the relevant setting in the workplace/ machine configuration

**Please note**

This processing is only relevant for "MDE workplaces" (see above).

If combined logon of order (see above) is active, then this option must be set to "N"

Serial number obligation

This identifier defines whether an order requires a serial number, and if so, how the serial number should be entered.

- N No serial number requirement
- + (plus) Serial number obligation, "positive" entry, i.e. the yield is entered with serial numbers.
- (minus) Serial number obligation, "negative" entry, i.e. the scrap or problem quantities are entered with serial numbers.

**Please note:**

The identifier "Serial number obligation" determines at the processing code or the operation directly as of which operation the serial number must be entered.

Assignment of serial numbers

This identifier is only relevant if the identifier "Serial number obligation" is not set to "N". It defines how serial numbers are allocated

- P Serial numbers are transferred/ expected from the PPS system.
- H Serial numbers are generated by HYDRA according to the number range configuration

**Please note:**

The "H" option is provided for future upgrades/modifications. Currently, only the "P" option is supported.

Order is not logged off automatically

Processing can take place subject to the termination of the order (header). Using this identifier can prevent the order from being set to finished if the last recordable operation was finished.

- N Finishing all recordable OPs automatically sets the order to finished (setting order status using the control indicator "E")
- J Order is not logged off automatically, i.e. finishing all recordable OPs does *not*

automatically set the order to finished.



Please note

Currently, only the deletion of order backlog data or the transfer of order backlog data into the long-term tables depends on the status of the order (header).

This option must be set for the GKP and GKM order types (order types for waiting period orders). Reason: the waiting period operations are all identified as "unrecordable". If the status of a waiting period operation were changed, or if a waiting period operation were deleted, this would result in the status of the order itself being set with control indicator "E". Conversely, this means that a waiting period order (header) must be explicitly set to finished ("Change status" function of MOC).

Milestone processing only for the last operation

Activation of milestone processing

Show OP info when logging on

Defines whether or not the note or the additional information is to be displayed when logging the
OP on to the Windows terminal (CTWIN, AIP).

			→→
--	--	--	----

Plausibility index tab

Check for specifications in backlog of orders

This identifier activates the following validation check during operation logon:

- | | |
|---|---|
| J | <p>An OP logon is only allowed at the planned workplace</p> <p>If this option is set, then during logon the system checks the workplace number entered at the terminal and compares it to the workplace number in the HYDRA order backlog for plausibility. An operation may only be logged on to the planned workplace. If an operation is to be produced at a different workplace (e.g. in the event a machine breaks down), this operation must be re-planned in graphic shop floor scheduling (HLS) or in order sequencing.</p> |
| G | <p>An operation logon is only allowed to the planned group, i.e. an operation logon is possible either to the planned workplace, to a workplace belonging to the group or to a workplace planned for the group (still in pool of groups).</p> |
| K | <p>An operation logon is possible either to the planned workplace or to a</p> |

workplace of the same category.

Please note: This check requires that a category is defined at both workplaces (not empty), i.e. at the workplace where the operation logon was planned AND at the workplace where the operation is in fact logged on. Furthermore, the operation to be logged on must already be planned for a workplace (no longer in the pool of groups).

N No validation check

Check status of preceding OP

This identifier activates the check of whether or when the current operation may be logged on.

S Checking status of preceding operation.
In this case, the status of the preceding operation must have the identifier "Successor can be logged on" set.

N No check

Check minimum send-ahead qty. of preceding OP

This identifier defines whether a validation check for the minimum send-ahead quantity should be run during operation logon.

M Check for minimum send-ahead quantity during operation logon. In this case, a minimum send-ahead quantity (yield) must have been posted to the preceding operation so that the succeeding operation can be logged on. The minimum send-ahead quantity is defined at the preceding operation.

N No check.



The relevant configuration needs to be enabled at the order type to check the minimum send ahead quantity, when logging OPs on. The system supports overlapping operations with respect to the minimum send ahead quantity (or the lead time). This is enabled at the processing code.

The check for the minimum send ahead quantity only considers the yield that has so far been collected (primary quantity unit).

Quantities are not converted. For this reason, please pay attention to the fact that adjacent operations have the same primary quantity unit.

Example:

Operation 0100	Target quantity 1000	Send ahead quantity 50
Operation 0200	Target quantity 1000	

If checking is enabled, operation 0200 can only be logged on as soon as a yield (in primary quantity unit) of at least 50 has been uploaded/confirmed

for operation 0100.



Checking does not include the operation status of the preceding operation. The current operation cannot be logged on, in case the preceding operation has already been finished, but the send-ahead quantity has not yet been reached.

Only OPs of the same order type can be logged on simultaneously

If this option is set, only operations that are assigned to the same order type may be logged on simultaneously.

This check will be performed if this option is set at least for one operation, i.e. the operation to be logged on or the operations that have already been logged on.

Check that shop floor papers are printed

This setting can activate a validation check during login of operations depending on the print status of the time tickets. Under the proper conditions, this function can help prevent erroneous postings or rather can "guide" production.

J An OP may only be logged on if its shop floor papers were printed beforehand.

N No validation check



Please note:

The print status is registered by means of a corresponding identifier at the operation. It can be set when printing the time tickets (additional license).

Check against M/O relation when person logs on

The *maximum number of employees needed to produce this operation (M/O relation, production)* can be defined in the operation data in the system.

Validation check when person logs on relating to:

J Machine/ operator relation. When a person logs on to an operation, a check is run to determine if the maximum number of employees has already been reached. If this is the case, this personnel logon is rejected.

N No check

K Component list of requirements to be transferred (planned, currently not used)



Please note:

The check is not performed at group workplaces.

If the M/O ratio defined at the operation is not an integer, the value is rounded up to the next whole number.

Advance logons of staff are also taken into account during this check.

Quantity check of send-ahead quantity

If this identifier is set, then a check is run for manual quantity postings at the terminal to determine whether the sum total of the previously entered *yield* and the currently entered yield is greater than the already produced yield of the preceding OP. If it is the case, the posting is not allowed with the yield quantity entered.

If this flag is not set, no validation check is run.



Please note:

This requires that the current operation and the preceding operation each have the same primary quantity unit.

Upload index tab

Upload order postings

Upload

It can be decided based on this identifier whether entered data (postings) of orders of this type must be uploaded to the ERP/PPS system or not.

- | | |
|---|---|
| J | Postings are uploaded. Which postings are uploaded and/or how they are uploaded is defined by the identifiers listed below. |
| N | Postings are not uploaded. In this case, the following identifiers are irrelevant. |

OP logins

If this identifier is set, operation logins (record type "A" postings) are uploaded to the ERP/PPS system. The same structure is maintained during an upload as for postings, but in this case the quantity and duration fields are set to 0.

Please note: When uploading to SAP via PP-PDC interface, the OP logins are assigned the SAP record type L10.

Postings of type "A" records are generated for each operation login, including those automatic logins at the time of automatic shift change process.

Partial confirmations/uploads

If this identifier is set, partial quantities (record type "T" postings) are uploaded to the ERP/PPS system. In this case, the posting generates U/E records that do not contain any quantities. Only the T-records still contain quantities.

As far as the ERP/PPS system is concerned, uploads based on T-records are displayed at the upload interface in the same manner as record type "U" posting records. However, they only contain quantities, no durations. Durations continue to be uploaded from the postings of record types "U" and/or "E".

If the identifier is not set, the U/E records generated by the posting process are, in addition to the durations, also given the accumulated quantities of all T records generated between logon and interruption/ logoff.

Upload of scrap including reason

Unlike with "Upload of partial confirmations", with "Upload of scrap including reasons" only the scrap from the partial confirmations is uploaded. Yield, rework quantity and problem quantity are transferred from the postings of record type U and/or E.

Please note: If both options are set, the "Partial confirmations" option has the higher priority.

Approved order postings only

If this option is set, order postings are not uploaded until they are approved. After five days at the latest, not even approved postings are uploaded (default setting, can be changed by calling up the parameter /ABZEICH= in the program myerprck.exe/.out).

Upload company of actual workplace

<HELP missing>

Order postings instead of batch postings

For MPL machines, batch postings are uploaded by default (record type "H" postings). Instead of batch postings, this option makes it possible to upload order postings (record type "U"/"E" postings).

Upload type

Reserved:

Change after upload

This identifier defines how changes/ corrections to postings uploaded to the ERP/PPS system are handled. The identifier is evaluated during posting maintenance as well as during event maintenance.

- | | |
|---|---|
| J | Changes/ corrections are allowed to already uploaded postings; changes are also uploaded to the PPS system. |
| E | Changes/ corrections are allowed to already uploaded postings; however, these corrections are not uploaded to the PPS system. |
| N | No more changes/ corrections are allowed to already uploaded postings |

**Please note:**

In posting maintenance, the identifier is evaluated directly in relation to the corresponding order. In event maintenance, on the other hand, the last order with an "N" or "E" applies, and all previous events are locked irrespective of the particular order type.

This option may not be set for customers who use the SAP PP-PDC interface, because it does not support cancellations/ corrections. Please contact MPDV to discuss options available to nonetheless upload corrections to SAP.

Cancellation message job end first

<HELP missing>

Upload of waiting period postings

If this identifier is set, then postings identified as "waiting period" in the posting (ade_protokoll.karenz = 'P' or 'M') are uploaded to the PPS system.

If only approved waiting period postings are uploaded, then the option "Upload approved order postings only" or "Upload approved personnel postings only" must be set.

Upload of batch logon postings

Please note: This option is approved for uploads using the HYDRA HY72PPS interface only and may not be set in conjunction with the PP-PDC, KK3, KK4 or HY71PPS interfaces!

This option causes batch logons to be uploaded via the HY72PPS upload interface. The upload record contains no quantities/ durations and the logon/ logoff times correspond to the posting time.



Please note: In HYDRA standard processing, order logons and batch logons are uploaded first, i.e. if an order is logged on twice and interrupted once between two posting intervals, then the two logons are uploaded before the interruption.

Personnel postings**Personnel postings**

If this identifier is set, then record type "B" postings (personnel postings) are uploaded to the PPS system.

**Please note:**

This option may not be set for customers who use the SAP PP-PDC

interface.

Upload personnel postings only

Upload personnel postings only

Approved personnel postings only

If this option is set, record type "B" postings (personnel postings and personnel waiting period postings, if configured for uploading) are not uploaded until after approval. After five days at the latest, not even approved postings are uploaded (default setting, can be changed by calling up the parameter /ABZEICH= in the program myerprck.exe/.out).

Upload of planning changes in shop floor scheduling

Upload PPS operations only

Upload PPS operations only

Upload delayed operations only

Upload delayed operations only

Upload operations with deviating assignment only

Upload operations with deviating assignment only

Quality index tab

CAQ area

This field is reserved for future functions.

Generate CAQ inspection orders

This field is reserved for future functions.

Options index tab

Priority

control

- | | |
|---|---|
| U | Priority control is order-related, i.e. the priority is set in the order (header) and has an impact on all OPs. |
| G | Priority control is operation-related. |

Priority management

- | | |
|---|--|
| J | Orders of this order type are considered during priority management (see Order group configuration). That is, when newly creating or manually modifying an order in HYDRA, the system runs a check to determine whether the assigned priority is allowed, and if the maximum number of priorities has been exceeded, the action is rejected. |
|---|--|

The priority check is only performed for order/ operation modifications if the

priority was actually changed during the modification process.

If the order is transferred by the PPS interface and the maximum number of priorities has been exceeded, the order will not be rejected as a result of this check. In this case, however, the priority will be automatically set to 1.

Please note

This check is only performed if the option *Priorities* is set to 'U'.

Only those orders are considered that have an order status that is set to the identifier "Priority check".

N Orders of this order type are not considered during priority management.

Order postings need to be signed

This option indicates that order postings for operations with this order type must be signed (also see the manual dealing with escalations "ANR.UNCERTIFIED_BOOKINGS").

Personnel postings need to be signed

This option indicates that personnel postings for operations with this order type must be signed (also see the manual dealing with escalations "ANR.UNCERTIFIED_BOOKINGS").

Order type for composition

Charging orders are taken into account by composition. Charging orders are assigned to melting orders. In this case, the planned order number of the charging order is the order number of the melting order.

"C" = Charging order,

"M" = Melting order

User Fields Index Tab

The system can be configured to enable the user fields for the object type "AUART".

17 Order Status Texts

Summary

Menu	Master data → Order → Order status texts
Transaction code	Osst
Function authorization	Mdostt

The order status text function allows you to configure the description for each order status.

Usage

In the status text dialog, all possible order statuses are designated with a descriptive text. These descriptions are then used in the status assignment performed later. The goal is to use standard status texts for all order types.

Procedure

Create the status texts that you would like to use for status assignment later.

Assign a unique status text number and an informative text.

Then, categorize each of the status texts to an order or operation status.

Integration

The texts defined here are displayed on the client (MOC) based on the order status assignment performed in various evaluations/reports.

Requirement



You need the customizing key in order to configure the order status texts. Settings relating to a customer specifically are configured in the customer specific documentation (CID).

Selection criteria

The application provides the following selection criteria:

Status text designation

Search for a status text by status text designation. You can also run a search using wildcards.

Field description

Field descriptions status text

Status text identification

Designation

Status text

18 Order Status Assignment

Summary

Menu	Master data → Order → Order status assignment
Transaction code	ost
Function authorization	mdost

Usage

The different order statuses can be configured for the various order types in the order status assignment function.

The order status defines the order's current status.

Integration

Because order data are recorded in HYDRA based on the operation, HYDRA manages one status for each individual operation. The status indicates whether the operation has not yet begun, for example, or whether it has begun, was interrupted or has already been finished.

In addition, a status is also kept in the order header, which provides information on whether the order has not yet begun, whether it is currently being processed or whether it is already finished. At the time the last operation that can be posted is finished, the status is set to finished in the order header. Operations that cannot be posted will not be automatically finished when this occurs.

Both order and operation statuses can be configured. To do so, the status texts must be defined first. Then, the status texts are assigned to the individual statuses. What must be kept in mind is that for each order type, the order status or the operation status must be configured with one control indicator each (exception: statuses with a control indicator "S" may exist multiple times).

All possible order and operation statuses are configured in status assignment and are assigned to status texts using unique status numbers. The selection dialog allows you to select and display statuses that have already been assigned to an order type.

Requirement



You need the customizing key in order to configure the order status assignment. Settings relating to a customer specifically are configured as described in the customer specific documentation (CID).



The order status texts and order types must be created first.

Selection criteria

Status type

Selection of the assigned status type

Status text

Defined status text

Order type

Order type that was assigned to the status

Field descriptions

This dialog is used to define the statuses used in both the order (header) as well as in the separate operations. There is also another option that allows you to define an operation in relation to the secondary status. To differentiate which identifier is relevant for which status type, the corresponding status type is written in parenthesis after the field:

- A = Order status
- G = Operation status
- S = Secondary status

Order type [A G S]

The status is configured based on the order type. This means that for each order type the relevant order and operation statuses must be defined. Enter a valid order type in this field, for which the status should apply.

Status type (A/G/S)

A	Status refers to the (entire) order
G	Status refers to the operation
S	This is the preparation status (=secondary status) at the operation.

Status (A/G/S)

This is the externally valid status.

Status index tab

Status text no. (A, G, S)

Reference to the order status text table

Symbol (G)

Symbol assigned to the status. It is displayed in different functions/ evaluations (e.g. order overview).

The following values are allowed (case sensitive):

Color of the LED	Value that needs to be entered in the editing dialog.	Value shown in the detail panel.
Light green	l.bmp	LED_LIGHT_GREEN
Dark green	e.bmp	LED_GREEN
Gray	v.bmp	LED_GREY
Yellow	u.bmp	LED_YELLOW
Yellow/ light green	f.bmp	LED_YELLOW_GREEN
Pink	n.bmp	LED_PINK
Black	a.bmp	LED_BLACK
Blue	p.bmp	LED_BLUE
Red	x.bmp	LED_RED

Color (G)

Color of the relevant status (text). At this time, this color is only used in HYDRA shop floor scheduling (HLS).

Entry index tab

Control

Status type A (order status)

V	Prepared Release (for automatic release)
L	In process At least one OP was started
E	Finished All OPs are finished or deleted.
S	None All others



Status type G (OP status)

V	Prepared "Release indicator" for status; before OP is edited for the first time
L	Running Is set automatically once OP is logged on
U	Interrupted Is set automatically once OP is interrupted
F	Autom. interrupted Is set by the server if you want the OP to be interrupted automatically at the time of a shift change (when using the MDE function at the terminal)
E	Finished Is set automatically once OP is finished
S	None Other status; not automatically set. Means that no work has yet been done on the OP

Please note

An OP has not yet begun if its status is set with a control indicator: "V" or "S".

Explicitly setting an order to the status using the control indicator "E" will result in all OPs being set to the status with the control indicator "E".

The control indicator "A" means "archived"; it is not set via configuration, but is instead a fixed setting made at the time the data are transferred into the long-term table.

Can be logged on

Check to assure if the CURRENT status has set this indicator. If the operation is interrupted/ finished, the field is ignored. .

Please note: If an OP or an order have been set with the flag "locked", then the OP may not be logged on under any circumstances (irrespective of this setting)

Sequencing list

If this status is set, the operation is shown in the sequencing list at the terminal.

Please note: if an OP or an order has been set with the flag "locked", then the OP may not be displayed under any circumstances (irrespective of this setting).

Successor can be logged on

If the indicator "Check of preceding OP" ("Plausibilities" index tab) is set to "S" = status at the order type, then this indicator is used to check whether the subsequent OP can be logged on.

The indicator relates to the preceding OP!

If, for example, an OP logon of the subsequent OP is not to be allowed until the preceding OP has at least begun, then this indicator must be set if the status is "running", "interrupted", "finished" (possibly "deleted").

If, for example, an OP logon of the subsequent OP is not to be allowed until the preceding OP has finished, then this indicator needs only to be set if the status is "finished" (possibly "deleted").

All preceding OPs resulting from relationships in ADE_AUFTRAGSNETZ, are checked in this regard.

Successor can be logged off

If the indicator "Can be finished" ("Entry" index tab) is set to "S" = status at the order type, then this indicator is used to check whether the subsequent OP can be logged off.

The indicator relates to the preceding OP!

Example 1: If an OP logoff of the subsequent OP is not to be allowed until the preceding OP has also *at least begun*, then this indicator must be set to "J" (y – yes) if the status is "running", "interrupted" or "finished".

Example 2: If an OP logoff of the subsequent OP is not to be allowed until the preceding OP has *finished*, then this indicator needs only to be set if the status is "finished".

All preceding OPs resulting from relationships in ADE_AUFTRAGSNETZ, are checked in this regard.

Release of subsequent OP

If this indicator is set, then setting the status will result in the status of the subsequent OP(s) being set to the release status, i.e. set to the status assigned with the control indicator "V".

This only happens if the subsequent OP(s) has (have) not yet been run, was (were) interrupted or has (have) been finished.

Thus, this option only makes sense if new operations are not created/ transferred using the status with the control indicator "V".

Please note: As opposed to processing up to ADE 7.1, the release is no longer given as a result of automatically scheduling the operation to a planned workplace, but instead only by setting the aforementioned release status.

Planning index tab

Planning

N	No planning (no transfer to the planning component)
T	Scheduling only
F	Scheduling and detailed planning (dispatching) - also includes the simulation



Please note:

The indicator "N" at the order overrides the indicator at the operations. Conversely, the indicators at the operations override the indicator "J" (y – yes) at the order

Posting index tab

Posting order duration

Posting

M	Posting to the RPA of the workplace status A posting to the RPA of the operation is made using the RPA of the status currently active at the workplace (default).
---	--

Posting RPA

Reserved; currently not used

Options index tab

Initial status

This relates to the initial status that should be set at the time an order (A) or an OP (G, S) is set.

Exactly one status must be set using this indicator for both the order as well as the operation.

Please note: If the initial status is not the status with the control indicator "V", then the status of the first operation as well as that of the order is set using the control indicator "V" after the first scheduling, provided that the indicator "Scheduling without implicit order release" is *not* set at the order type.

Status can be set manually

Reserved; currently not used.

Authorization

Reserved; currently not used

Change of secondary status

Change of secondary status allowed (J (y – yes)/N)

Priority check PRIO_PRUEFUNG

Check the priority (J (Y – yes)/N)

When an order is newly created/ modified, the system checks it against priority management (order group). For this evaluation, all orders with statuses assigned to this indicator are checked.

Please note: The priority check is only performed for order/ operation modifications if the priority was actually changed during the modification process.

Alterable order data

If this status is set, the order or the OP may either be altered by the PPS system or manually or it may no longer be altered.

J = MOC and MLE interface (PPS=J) alterable

N = MOC and MLE interface not alterable

K = Only MOC alterable (not via the MLE interface)

M = Only MLE interface alterable (not via the console)

This option does not refer to the priority, notes, long texts, secondary statuses.

Deletable

May the order or the operation be deleted in this status (manually, via PPS system)?

Order

Yes, BUT only if the operations can be deleted, i.e. at no operation may a status be set that has a control indicator "L" AND for which the "deletable" indicator is set to "N".

Operation

If the OP is running, then the deletion flag is set.

Please note: Operations with a current status with a control indicator "L" or "F" may as a rule not be deleted, irrespective of this setting.

Action

Behavior indicating how the order or the operation should respond during the deletion process, i.e. at the time of deletion (now) or when a deletion flag is triggered:

Order

- | | |
|---|--|
| L | If something is physically deleted and all operations are physically deleted as well (order header controls the behavior) |
| E | Order is set to the status with the control indicator "E".
WARNING: Only the order header, i.e. for the OPs the indicator must also be set! |
| D | Order is set to status with control indicator "D" and is deleted/ archived using the archiving/ deletion program (data management) after the set period. How |

the operations are treated is described via the indicator RESET ACTION (LOESCHAKTION) at the status of each OP.

- X If the action was triggered by the MLE interface:
Order is set to the status with the control indicator "E".
WARNING: Only the order header, i.e. for the OPs the indicator must also be set!
If the action was NOT triggered by the MLE interface:
If something is physically deleted and all operations are physically deleted as well (order header controls the behavior)

Operation

- L Is physically deleted.
- D Is set to the status with the control indicator "D".
- E OP is set to the status with the control indicator "E" (fixed).
- X If the action was triggered by the MLE interface:
OP is set to the status with the control indicator "E" (fixed).
If the action was NOT triggered by the MLE interface:
OP is physically deleted.



The reset/deletion action should be the same for both the order header as well as for the OPs, because currently the settings each relate to the order or to OPs and the OPs do not "inherit" anything from their order header.

To assure that an order or an operation can no longer be deleted as of a certain status, the indicator must be set at all corresponding statuses.

19 Order-Related Posting

Overview

	→	
	→→→	
Menu	Data collection → Corrections → Order-related postings	
Transaction code	oboo	
Function authorization	oboo	

Usage

The order-related postings application is ideal for any foreman, team leader, supervisor or anyone else who reviews/ adjusts postings entered during production.

Integration

The order-related postings represent a database of all of the data entered during production. More precisely, listed here are all postings and results from production and as such they are available to be reviewed and can be modified.

Displayed in the list are all postings matching the selection made in the selection panel. Displayed are only postings for which the user is authorized accordingly to the responsibility area for the workplace/ machine (record types not equal to "B") or for the persons (record type "B") the postings are assigned to.

Requirement

Postings (logon, interruption, partial confirmation, logoff) must be entered for each operation.

The basic parameter setting of HYDRA "maintenance on the basis of events" determines the scope of editing postings in the maintenance of postings function.

If this option is *not* set all postings may be edited in the maintenance of postings function, i.e. postings recorded manually using the maintenance of postings function as well as postings generated automatically by HYDRA due to the recorded events.

However, in case this option is enabled postings generated automatically in HYDRA (due to events) can neither be corrected nor deleted in the maintenance of postings function. Such postings can only be edited by way of the "event maintenance" function. The maintenance of postings function only allows for the manually recorded postings to be corrected or deleted.

Please note:

MPDV recommends using the maintenance of postings function. This option is set by default for new installations.

Automatically generated postings and manually entered postings

HYDRA distinguishes between automatically generated postings and manually recorded postings. Automatically generated postings are postings generated automatically by HYDRA based on the recorded posting events. In contrast to this, manually recorded postings are postings entered manually in the maintenance of postings function. They do not relate to posting events.



HYDRA does neither generate nor correct subsequent events for manually created or edited postings in the maintenance of postings dialog!

If a BDE (shop floor data collection) posting is changed/initially created or deleted does not have an impact on MDE (machine data collection) postings!

A modification/initial data creation or deletion does not result in a recalculation of the actual cycle relating to operations.

Selection criteria

The application provides the following selection criteria:

Order

All postings for this order are shown when the order number is entered.

Operation

This selection criterion refers to the operation number. All postings are shown that are posted to the operation number that was entered.

Workplace

This selection criterion relates to the workplace number where the posting originated. All postings are displayed that were generated as a result of the events at this machine/ this workplace.

Cost center

By entering a cost center, all postings are shown that were posted at the workplaces/ machines of this cost center in the period of time entered.

Record type

It is possible to limit the selection by record type ("U/ E/ H/ B/ T records"). What needs to be considered when doing so is that only postings of record type "H" are shown for operations that require batch management, unless there was a selection explicitly by record type.

Person from ... to ...

By entering a personnel number, all personnel postings ("B records") can be displayed that were generated from the personnel postings for the person entered as well as postings, which were generated with the staff badge number for postings entered manually.

Order type

By selecting one or more categories, all postings are displayed with an order that is assigned to this order type.

Order category

By selecting one or more categories, all postings are displayed with an order that is assigned to this category.

Interruption reason

By entering an interruption reason, all postings are displayed that were canceled or interrupted with this interruption reason.

Premium group

In HYDRA Incentive Wage Calculation LLE, either workplaces/ machines are assigned to premium groups or persons are assigned to premium groups. By entering a premium group, all postings are shown that were posted at the workplaces/ machines of this premium group in the period of time entered.

Modified by/ modified on

By entering data in the "modified by" field and the editing date, all postings will be displayed that a person edited on this day.

Including cancellation log records

This option is used to select and display cancellation log records (postings of record type "S").

Including original log records

Original postings (postings of record type "O") are also displayed from this option.

Non-authorized postings

Use this check box to display order-related postings (record type U, E, H, T, B) that have not yet been authorized. This makes sense if postings are not uploaded until they have been authorized.

Date from ... to ...

All of the postings made in the limited period are displayed.

Different results might occur for a date, depending on whether the selection made is based on the shift or time.



Please note that this application only allows for data to be selected and edited in the online data area.

Shift

By entering the shift, postings can be selected that were posted during a certain shift (as per the BDE shift model).

Please keep in mind that the time stamp for logging off pertaining to a BDE log record determines the shift (shift date, shift number) to which the log record belongs.

Time

Postings can be limited even further using the time selection criterion. This selection criterion references the posting's start time. Please note in this context that this start date may definitely go back farther into the past for workplaces where the shift automatic function has not been activated.

When using several selection criteria, the overlapping selection criteria is displayed.



Please note as concerns merged operations: For merged operations formed at the MOC, ONLY the individual operations are displayed and only these can be modified. When log records of individual workplaces are modified/ corrected, this will update the status of the merged operation.



Please note as concerns split operations: For split operations, the log records for the split operations are displayed. No log records are generated from the split operations for the split master. When log records of split operations are modified/ corrected, this will update the status of the split master.

Toolbar



Authorize

Function to [validate postings\archive\mast_ind\translate\across\sign - sign](#)



Machine-related postings

Starts the application [machine-related postings](#)

Transfers the workplace, cost center, date/time and the shift from the selection panel.



Order information

Opens the [order information](#)

Field description

In the table view, the posting's data, including any other additional information, is displayed, while in the maintenance/editing dialog only the data relevant to the posting are displayed.

General index tab

Record type

Record type of an order-related posting:

U: The posting was triggered by the operation being interrupted.

E: The posting was triggered by the operation being ended.

B: Personnel-related posting ("B" is for "Bediener" = operator)

H: Posting relating to batches: Posting is made with respect to batches for operations that are subject to batch management, provided that HYDRA-MPL is in use. Postings of the record type U or E are not displayed in this case if the record type is not selected explicitly.

T: Partial quantity confirmation/upload (scrap with reason)

Order

Order number

Sequence

If sequences are used, the sequence is displayed here.

OP

The order/ operation specified must exist; there is no review relating to the status.

Split

If splits are used, the split number is displayed here.

Workplace

Number of the workplace/ machine at which the operation was posted.

Cost center

Cost center of the workplace or cost center at which the posting was made

Person

Personnel number of the person who triggered the posting.

This number must exist in HYDRA. For personal postings (record type "B"), entering the personnel number is mandatory. The field can remain empty for another record type.

Employee group (only displayed in tabular form)

Assigned employee group as per HR master data

Interruption reason

Status of the machine/ workplace at the time of the interruption or the completion confirmation. This relates to the workplace/ machine status.

Batch number (only relevant when using MPL)

Produced batch number or entered batch number

Quantity index tab

Primary quantity

Yield

Yield produced in the primary quantity unit

Scrap

Scrap produced in the primary quantity unit

Scrap reason

Scrap reason

Rework

Rework quantity in the primary quantity unit

Open quantity

Open quantity entered in the primary quantity unit

Unit

Quantity unit, is assumed from the operation.

Please note: the field cannot be modified in editing mode

Total cycles (only displayed in tabular form)

Cycles posted while the operation was logged on (only for postings of record type U, E)

Cycles during MUT (only displayed in tabular form)

Cycles posted while the operation was logged on and for which the status "Production" was active
(only for postings of record type U, E)

Secondary quantity/ tertiary quantity/ basic quantity

Yield, scrap, rework, open quantity, unit

Posted quantities in each quantity unit.

Please note: the units fields cannot be modified in editing mode.

Duration index tab

Logon

Date and time of the start posting which this posting results from.

Logoff

Date and time of the start posting or the end posting which this posting results from.



The points in time "first logon", "last interruption" or "last logoff" saved in the order status are not affected by modifications to the points in time "logon" or "logoff".

Shift, shift date, beginning of shift, end of shift

Number of the shift, shift date that the posting is assigned to according to the BDE shift model as well as the shift times.

Shift type (only displayed in tabular form)

Shift type that was defined in the day type configuration.

Shift model (only displayed in tabular form)

Day type that was used for the posting.

Order-related RPA

Order-related times that were posted to the resource performance accounts.

Personal RPA

Personal times that were posted to the resource performance accounts.

Duration (only displayed in tabular form)

Posting duration.

Sum (in the editing dialog only)

The value corresponds to the duration of the posting (column "duration" in the tabular view) and is calculated automatically from the total of RPA 1 to RPA 11.

Labor utilization

Duration of labor utilization

Difference (in the editing dialog only)

The difference between the original duration and

- the new duration from the total of RPA1-11 or
- from the different posting times

is shown for information purposes.

In general, order-related resource performance accounts and personal resource performance accounts are displayed and may be edited separately.

For personal postings (record type "B"), order-related and personal postings are shown subject to the setup configuration option "proportionate RPA posting in personnel postings".

HYDRA basic parameter settings: option "proportionate RPA posting in personnel postings"

If the option "proportionate RPA posting in personnel postings" is enabled in the HYDRA basic parameter settings, the order-related as well as personal resource performance accounts will be displayed for personnel postings (record type "B").

If the option “proportionate RPA posting in personnel postings” is not enabled in the HYDRA basic parameter settings, only the personal resource performance accounts will be displayed for personal postings (record type “B”).

The order-related resource performance accounts are automatically set to the values of the personal resource performance accounts.

Field descriptions in the wage data index tab

te

Default individual time for labor time in [h/1000]. Is assumed from the operation (requires the relevant LLE license).

tr

Target specification for setup time in [h]. Is assumed from the operation (requires the relevant LLE license).

teb

Default individual time for the (machine) occupancy time in [h/1000]. Is assumed from the operation (requires the relevant LLE license).

trb

Default individual time for the (machine) setup time in [h]. Is assumed from the operation (requires the relevant LLE license).

Wage type

The wage type is assumed from the operation (requires the relevant LLE license). Please note: Not this value is uploaded as the wage type at the ERP/PPS upload interface, but the value defined in the order backlog at the operation.

Premium group

In HYDRA Incentive Wage Calculation LLE, either workplaces/ machines are assigned to premium groups or persons are assigned to premium groups (requires the relevant LLE license). In this case, the specified premium group must be valid.

Operator position

Operator position entered for the posting (only relevant for personnel postings).

Premium indicator

Premium indicator entered for the posting (only relevant for personnel postings).

Settlement date

The settlement date determines if the posting belongs to a settlement date in order to determine the incentive wage and to perform the PZE-BDE comparison. Once the settlement date has been filled in, then the order-related posting is assigned to this day. If the field *Settlement date* is empty, the field *Shift date* will be assigned. If the shift date frequently does not match the assignment to the PZE (time and attendance module), the BDE shift models will need to be adjusted.

The field is filled in as a result of the PZE work day evaluation or by the maintenance of postings. This means that normally the field is initially empty for all postings for the current day and that it will not be updated until the next morning after the PZE work day evaluation has run. As a result, postings for the current day may be incorrectly assigned to the previous day. The incorrect assignment will normally disappear then on the next day on its own.

Settlement date assignment is subject to the following rules:

Order-related personnel postings (B records)

Personnel postings are assigned to the PZE personal day if they extend into a time window of +/- 2 hours of the rounded working times of the reporting person. If a personnel posting lasts several days, it will be assigned to the first PZE personal day, because often a person has forgotten to log off.

Order-related postings (U/E/T records)

Order-related postings are assigned to the PZE personal day of the reporting person if their end time is within a time window of +/- 2 hours of the rounded working time of the reporting person. If a personal posting lasts several days, it will be assigned to the last PZE personal day, because the reporting person is determined based on the logoff and the person logging off posts the quantities. If no personnel number is entered in the posting, the settlement day cannot be assigned.

User fields index tab

All of the user fields are displayed with the key Object type= "ADEPRO" and User field key= "SYSTEM".

Administration index tab**Editor**

The last person to edit the posting.

Last modification

Date on which the posting was last edited.

Name

The editor's name who edited the posting at last according to the user administration.

Uploaded

A flag showing whether the posting has already been uploaded to an upper level system. J (Y) = uploaded, N = not uploaded, X = blocked

Upload date, upload time (only displayed in tabular form)

Time of the upload

Wage confirmed

Flag to upload data to a payroll system according to the customer's requirements.

Date of authorization, authorization time (only displayed in tabular form)

Date on which the posting was last authorized.

Reference

Unique reference for the posting

Type

This flag indicates how this data record was entered or modified.

(empty): Original posting from entry.

E: Manually created posting as a result of modifications. Postings that have already been edited (Type "E") are not copied and are stored as a type "O" posting.

O: Once an original posting has been edited, it is copied and stored as a type "O". Thus, original information will remain available to be displayed.

S: Cancellation for PPS

cancellation and original postings cannot be modified or deleted.

Create new posting

Along with the manual creation of a posting of the record type "U" or "E", a posting of the record type "T" is created automatically.

The posting of record type T includes quantities of the U/E posting, but no durations, the point in time refers to the end time of the U/E posting

Automatic status change of the operation

- If a posting of the record type "E" is created for an operation the operation will be set to the operation status "E", provided that this status was "V" or "U" previously. Additionally, the point in time of the "first login" or the "last logoff" are set in the order status, provided they had not yet been set.
- If a posting of the record type "U" is created for an operation, the operation will be set to the operation status "U", provided that this status was "V" previously. Additionally, the point in time of the "first login" or the "last interruption" are set in the order status, provided they had not yet been set.
- Please note: The mechanisms that are processed are not the same in both cases, in contrast to the interruption of an operation (A_UN) or the logging off of an operation (A_AB/A_BE).
- The points in time of the "first logon", "last interruption" or "last logoff" saved in the order status are not adjusted, in case the operation has already been started. In this case, the date corresponds to the date of the real, first logon.

Automatic quantity adjustment of the operation

- If a posting of the record type “E” or “U” is created for an operation the order quantity will be corrected in the order status. A partial upload (posting of the record type “T”) including the quantity is generated additionally
- If a posting of the record type “T” is created for an operation the order quantity will be adjusted in the corresponding order posting of the record type “U” or “E” and order quantities will be revised in the order status
- Target quantities will not be compared if a posting of the record type “E” is created

Change posting

A cancelation record will be generated if a posting is already uploaded to the PPS system (upload flag “J”). In any other case, the data record is changed physically.



Please note: At the time of changing a posting, ERP/PPS uploads must not be running.

Automatisms to change postings

- The original data record will be saved if the posting is generated automatically.
- Cancelation postings and original postings cannot be changed.

Changing of already uploaded postings

As a part of HYDRA customizing it can be defined for each order type whether or not the changes made in the posting maintenance are to be uploaded to the ERP/PPS system. The following possibilities are given:

- | | |
|---|---|
| J | Allow modification and upload |
| E | Allow modification; no upload – data may be changed but not uploaded. |
| N | No, modifications are not allowed – data can no longer be changed. |

Checking respectively refers to the order type of the order to be changed.

A cancelation posting is generated when changing a posting that has already been uploaded, provided that the option is set to “J”. This cancelation posting keeps the original reference. The changed posting is created with the new reference.

Please note: No check is performed for MDE postings.

Changing of the record type

The record type of a posting cannot be changed from

- record type "H" / "U" / "E" to record type "B"
- record type "B" to record type "H" / "U" / "E"

and is rejected by a validity check.

Correcting of an order number in a posting

The below-mentioned procedure/order has to be observed to change the order number in a posting:

- Copy the posting to be changed to the new order number.
Copy the order-related posting (record type "U", "E") at first and then the personal posting (record type "B").
- Delete the postings made for the wrong order number.

Changing the quantity for an order posting (record type "U"/"E")

- If the quantity is reduced a partial upload (posting of the record type "T") including the difference quantity will be generated. This posting includes negative quantities
- If the quantity is increased a partial upload (posting of the record type "T") including the difference quantity will be generated. In this case, this posting includes positive quantities
- The scrap reason from the U-posting is used as scrap reason for the new posting (record type "T")
- Order quantities are corrected in the order status
- Target quantities are not compared if a posting of the record type "E" is corrected.
- If the Material and Production Logistics module (HYDRA-MPL) is in use quantities may ONLY be corrected within the H records and not within the U/E postings!

Changing quantities for a partial upload (record type "T")

- The quantity is adjusted in the corresponding posting relating to the order (record type "U" or "E").
- The batch record (posting of the record type "H") is adjusted, provided that the operation is subject to management in batches

Changing quantities for a batch posting (record type "H")

- If the quantity is reduced a partial upload (posting of the record type "T") including the difference quantity will be generated. This posting includes negative quantities
- If the quantity is increased a partial upload (posting of the record type "T") including the difference quantity will be generated. In this case, this posting includes positive quantities
- The quantities are adjusted in the U/E record.
- The batch status is not adjusted!

- If the Material and Production Logistics module (HYDRA-MPL) is in use quantities may ONLY be corrected within the H records!

Changing quantities for a personal posting (record type "B")

Quantity changes to personal postings generally do not affect the order status or order-related postings.

Changing the time in an order posting (record type "U"/"E")

- If times (logon/logoff) of an order posting (record type "U" or "E") are changed the times for the partial upload(s) (record type "T") will also be changed, in case this upload/these uploads would then no longer be within the order posting.
- If the times (logon/logoff) of an order posting (record type "U" or "E") are to be shortened, the times of personnel postings (record type = "B") that might pertain to it need to be changed at first. As otherwise, there would be personnel postings outside of order postings and could no longer be changed.
- The points in time of the "first logon", "last interruption" or "last logoff" saved in the order status are no longer adjusted by changing the points in time (logon/logoff).

Changing the time in a personnel posting (record type "B")

- If points in time (logon/logoff) pertaining to a personnel posting (record type "B") are changed outside of an order posting, an error message will occur. As a personnel posting must not exist without a relevant order posting.

Procedure for changing times in posting records

- If an order posting is shortened, personnel postings that might pertain to it need to be shortened first.
- In case a personnel posting is to be prolonged (longer than the original order posting), the order posting that pertains to it needs to be prolonged at first.

Changing of RPA in personal postings (record type "B")

- The personal resource performance accounts are adjusted accordingly in the order posting. This is generally performed irrespective of the option "Proportionate RPA posting in staff postings" in the HYDRA basic parameter settings.
- If an RPA is changed the labor time is adjusted in the section "personal resource performance accounts". The "duration" field shows the sum total of RPAs. The total of RPAs may deviate from the labor time as labor times can be changed manually.

Labor time / personal RPAs in order-related postings (record type "U"/"E")

- In order-related postings the field "labor time" is not adjusted automatically if modifications to the personnel postings occur. The labor time needs to be changed manually within the order posting.

- If a personnel posting is changed, the personal resource performance accounts are adjusted accordingly within the order posting. This is generally performed, irrespective of the option "Proportionate RPA posting in staff postings" in the HYDRA basic parameter settings.

Copy posting

The copy function is the same as adding a new posting. In this case, though, the fields in the dialog box are preset to the values of the currently selected posting.

Delete posting

- If a posting is already confirmed to the PPS system (upload flag "J"), then a cancellation record is created. Otherwise, the data record is physically deleted.
- The operation is reactivated automatically if a posting of the record type "E" is deleted.
- When deleting a posting of the record type "U" or "E", all the partial uploads (record "T") pertaining to it will be deleted automatically.
- Personal postings (record type "B") are not deleted along with deleting order-related postings (record type "U"/"E"). Consequently, they should be deleted manually prior to deleting order-related postings.
- Only manually created postings can be deleted if the HYDRA basic parameter option "maintenance on the basis of events" is enabled.
- The saved points in time "first logon", "last interruption" or "last logoff" are not updated in the order status if an order-related posting is deleted (record type "U"/"E").

Authorizing of postings

The date of the approval as well as the person giving the approval are set during the authorization process.

Please note

When HYDRA is customized, it is possible to define for individual order types, that postings can only be uploaded/confirmed after authorization.

After 5 days at the latest (this standard setting can be changed) non-authorized postings are also confirmed/uploaded.

General checking

- In case a posting has already been uploaded and the customizing option "no change after upload" is set, the posting can neither be changed nor deleted anymore.

- In case a posting has already been uploaded and the configuration “change allowed, no upload” is enabled in the order type dialog, the posting can be changed/deleted after confirming it in an information dialog. In this case the status is corrected in HYDRA but an upload/cancelation is not sent to the higher-level system.
- Postings entered/generated automatically can only be edited or deleted if the event maintenance function is disabled. In contrast to this, posting records created manually may always be changed or deleted.
- Cancelation postings (record type “S”) and original postings (record type “O”) cannot be changed/deleted, as these postings have been designed for transparency/traceability reasons.
- When order postings (U/E records) are added/changed/copied, it is checked whether they coincide with already existing order postings. The period, order number and machine are checked in this context.
- When a partial upload (record type “T”) is added/changed/copied, it is checked whether a corresponding order posting exists.
- When a batch posting (record type “H”) is added/changed/copied, it is checked whether
 - a corresponding order posting exists
 - the operation is subject to management in batches
 - any overlapping batch posting exists for the order/workplace.
 - only yield or scrap is indicated. But not both quantity units.
- The user must be provided with the responsibility area authorization for the workplace indicated in the posting record.
- Uploads must not be running when a posting is being changed
- When the posting maintenance functionality is used to change automatically collected data and to create completely new postings (e.g. external orders), it is checked whether or not the workplace is currently blocked by the event maintenance function.
- →→→→

20 Event Maintenance

Summary

Menu	Data entry → Data correction → Event maintenance
Transaction code	evli
Function authorization	evli

Usage

Event maintenance offers the ability to edit and recalculate posting events resulting from data entry. Furthermore, events can be deleted or new events created.

Integration

The update function gives the operator the ability to subsequently modify events and to have these modifications recalculated by the system after a corresponding validation check and in so doing, to obtain modified postings that can be taken into consideration (selected) in the various evaluations/reports and/or uploaded to the higher-level ERP system.

The distinctive feature of event maintenance, as opposed to posting maintenance, is that it has a recalculation function that makes it possible to regenerate postings that result from the modified data (events). In the process, like on-line posting at the time of data entry, all validation checks necessary for generating a consistent database are likewise performed.

Requirements

The option needs to be activated to be able to edit events in this application. If this is not the case, the following note will be displayed, when opening the application “The event-related BDE maintenance has been disabled in the basic parameter settings. Modifications cannot be saved.” Consequently, events can only be selected and displayed. But new events cannot be added. Existing events can neither be edited nor deleted.

The maintenance of events option is disabled by default. The option is only enabled if required as a part of customizing the system and once the corresponding prerequisites or the below-mentioned restrictions have been checked and discussed with the customer.

Restrictions

Certain posting functions are purely data entry functions and the data entered using them cannot be edited or recalculated in event maintenance. Use of the tabular event maintenance in conjunction with these functions must be tested in each case separately. These functions are described below.

Uneditable data that has been entered

- Remarks or comments entered in different posting dialogs at the shop floor terminal cannot be displayed or edited in event maintenance.
- Serial numbers collected for OPs requiring serial numbers cannot be displayed or edited in event maintenance.
- Quantities can be entered in the system, including in alternative units. However, in event maintenance, quantities can only be edited in their primary quantity unit.
- Manual activities can be entered in the system or calculated based on formulas. Activities cannot be displayed or edited in event maintenance.
- Posting using OP reference or upload number:
Editing can only be done in event maintenance based on the MES order number.
- Production lock modifications are documented as events . They can be displayed in event maintenance, but not edited.
- User fields that during the customizing process were integrated into posting dialogs (e.g. OP logoff) cannot be displayed or edited in event maintenance.
- Automatic posting processes and their configurations (e.g. terminating predecessors/ interrupting or automatically logging off an operation if its target quantity is reached), are not taken into consideration during recalculation in event maintenance.
- Events that occur because of resource-based postings (WRM or DNC) are not displayed or edited in events maintenance.
- The "Automatic OP change" option in the *BDE* index tab of HR master data is only taken into account online; in event maintenance it does not affect recalculation.
- The "OP change during status change" option in status assignment of machines/ workplaces is only taken into account online; in event maintenance, it does not affect recalculation.
- Postings of personnel or machine waiting periods are not displayed in events maintenance.
- No escalations are triggered during recalculation in events maintenance.
- Additionally recorded optional data in the "Change machine status" dialog is not displayed in the input dialog and cannot be modified. This refers to data such as "expected duration", a comment or any additional data for escalation management.

Merged operations

Assignment of a new OP to a merged operation is NOT supported in event maintenance. Only data for already posted merged operations can be modified.

Merged operations that were entered using the "Merged operation per machine" function CANNOT be edited in events maintenance.

A recalculation is no longer possible for merged operations that were generated in MOC if the composition of the MOP was changed.

PZE controls ADE

The "PZE controls ADE" setup setting can lead to scenarios in event maintenance in which the personnel logon event is displayed before the actual order logon, because the logon is then rounded to the PZE time.

If inconsistent postings exist in event maintenance, evident because a personnel logon appears before the actual order logon, then they must be corrected so that the personnel logon time is changed to the time of the order logon.

Interface with SAP

Event maintenance cannot be used in conjunction with the PP-PDC interface, because the standard SAP system does not permit cancellations from external systems.

MDE change of shifts events

Changing these events might result in gaps or the shifts being no longer calculated completely. For this reason, shift change events should not be modified in general.

In this case, recalculation of the entire shift might not be guaranteed, as the relevant events identifying the points in time of the shift start/shift end can no longer be found at the corresponding point in time.

Change of shift events are events triggering status changes "M_MST"

end of shift → dialog A_AUN

beginning of shift → dialog A_AAN

Shift end events (event M_MST with dialog A_AUN) only may include the status 20000 "no shift".

Editing of finished operations

Operations that have already been finished can be edited in the event maintenance dialog. However, please note in this context, that the status of a finished operation does no longer change, not even if, for example, the logoff of the operation is canceled or changed into an interruption.

Operations that have already been finished generally have to be reactivated using the function "reactivate operation".

Material and production logistic (MPL)

There are the following restrictions for the event maintenance in conjunction with operations subject to management in batches:

- Only MPL machines with batch processing can be edited. Machines of the types cutting reels, parallel output batches, throughput batch mode, etc. are not supported.

- Input batch postings relating to the operation are only supported to a limited extent for editing. Postings outside of the OP posting are not taken into account. Consequently, consumption postings to input batches are not considered.
- It is not allowed to change output batch numbers or to delete events CA_AN/CA_AB.
- Batch attributes can neither be displayed nor changed using the event maintenance.
- Existing batch tracing is not changed subsequently by the editing function.

Editor of the event

For technical reasons, a user ID that has a maximum of 9 characters can be defined for the event (column "editor"). This means, inputting a user ID that is 10 characters long results in the last digit to be cut off in the "editor" column of event maintenance.

Edit partitioning

Editing of the machine partitioning event (M_TLG) only affects the shift included in recalculation.

If an order runs over several shifts (automatic relogin), all events affected by this event changing the partitioning need to be recalculated.

For this reason, the following structure is to be kept for changing the partitioning event:

- If partitioning is changed, the entire period for which this modification applies needs to be requested, i.e. until the operation is interrupted or finished manually.
- When changes are made, the relevant partitioning event has to be changed

AND

if the operation runs automatically over several shifts, the subsequent manual interruption or completion confirmation of the order has to be changed as well.

For the manual interruption/completion confirmation of the order, it is sufficient to just save the data without changing it.

- Consequently, the recalculation process automatically recalculates all data that coincide with the period between changing the machine partitioning and the manual interruption/completion conformation.

Please note:

The events to be recalculated are not selected automatically, as changes to the partitioning might affect several shifts that might then no longer be included in the selection period.

Selection criteria

The application provides the following selection criteria:

Workplace

All data relating to the selected workplace are shown.

MES order number

Results list: The workplace events, to which the order was posted during the selected period, are shown. You can also run a search using wildcards.

Person ... to ...

Results list: The workplace events the person (persons) was (were) logged onto during the selected period, are displayed. You cannot run a search using wildcards.

Date ... to ...

The period of time within which the events should be selected and displayed.



Please note that this application only allows for data to be selected and edited in the online data area.

Parallel staff logins

The "Parallel staff logins" option selects the data for all workplaces a person was logged onto. It should be set if data is being edited on workplaces where staff is posted for multiple machine operation.

More information is available [here](#).

Refresh data

This option allows you to control whether data in event maintenance should be refreshed when recalculation is completed (slower performance in the event of large amounts of data). Please note: In the event of a recalculation error, the data freezes in the display.



Except for the check box "Refresh data", all selection fields are disabled when data is requested. They cannot be input again until you have clicked on the "Discard" icon.

If the data are currently locked by another editor at the time data are requested, then the user will still have the option to have the data be displayed. In this case, the data are only displayed;

the data may not be modified.

Toolbar

The following functions can be accessed from the toolbar. Please also consider the [requirements](#) for editing events:



Modify an existing event

Function authorization: evli.edit

The dialog opens that corresponds to the event selected.



Delete an existing event

Function authorization: evli.delete

The selected event is deleted.



Recalculate

After acknowledging the confirmation prompt, any added, modified or deleted events are checked for validity; the postings resulting from the events are regenerated. Once recalculation is complete, a note accordingly will finish the action.

If an error occurs during recalculation (e.g. because of logically invalid values), then this is indicated by a message.



Discard

Any added, modified or deleted events are discarded after acknowledgment by the user and the list is cleared.



Order information

Function authorization: orin

Calls up the application [order information](#).



Workplaces/ machines

Function authorization: wpov

Calls up the application [workplaces/machines](#).

Create events index tab

Function authorization: evli.create

The icons  for creating events are available in this index tab. The icons are divided into separate categories:

Operation category:

- Log OP on (A_AN)
- Log OP off (A_AB)
- Partial confirmation (A_TR)
- Quantity upload (A_MR)
- Interrupt OP (A_UN)
- Finish OP (A_BE)

Person category:

- Log person on (P_AN)
- Log person off (P_AB)

Machine/ workplace category:

- Change status (M_MST)
- Change partitioning (M_TLG),
- Automatic counter (M_CTR_AUTO),
- Automatic quantity (M_AUTO)

Materials category:

- Log output batch on (CA_AN)
- Log output batch off (CA_AB)
- Log input batch on (CE_AN)
- Log input batch off (CE_AB)

Miscellaneous category:

- Activate OP (NC_AN)
- Deactivate OP (NC_AB)

• →→→→**Field descriptions**

Class

Internal event classification:

P = Personal data

M = Workplace data

A = Order data

C = Batch data

Event

The possible events and their colors are listed [here](#).

Dialog

Each event-triggered dialog is displayed here. Besides the dialogs of the same name (e.g. A_AN, A_UN, ...) a different dialog can be the trigger for the event.

Date, time

The point in time when the event was posted.

Workplace

The workplace, to which the event was posted.

MES order number

The combined order/ operation number, for which the event was posted. This field is only filled in if the event has an order reference.

Person

The person who triggered the event or for whom the event was triggered. This field is only filled in if the event was triggered by a person or has a reference to a person.

Status

The workplace/ machine status that applied at the time of the event; this field contains only context-related information.

PZE

Reference to clock-in or clock-out

Primary quantity

Yield, yield reason, scrap, scrap reason, rework quantity, rework quantity reason, open quantity, open quantity reason

Secondary quantity

Yield, yield reason, scrap, scrap reason, rework quantity, rework quantity reason, open quantity, open quantity reason

Tertiary quantity

Yield, yield reason, scrap, scrap reason, rework quantity, rework quantity reason, open quantity, open quantity reason

Basic quantity

Yield, yield reason, scrap, scrap reason, rework quantity, rework quantity reason, open quantity, open quantity reason

Editor

Last editor of the event

Date, time

Time of the last edit

Editable

Y = Event may be edited

N = Event may not be edited, e.g. is the master of an MOP or is locked by a change in the log record

Reference

Unique ID of the data record

Priority

Priority specification for simultaneous events (1 = highest priority)

Sorting

Internal use

Status 1

Event M_MST: machine status/ interruption reason

Event M_TLG: Partitioning

Event M_SZY: Target cycle

Status 2

Internal use

Status 3

Internal use

Attribute 1

Events C_GEN, C_UMB, CE_AN, CE_AB, CA_AN, CA_AB: Target location/ material buffer

Attribute 2

Event P_AN: Wage/ premium indicator

Events CE_AB, C_GEN: Info on batch

Otherwise: Internal use

Attribute 3

Event P_AN: Operator position

Events CA_AB, C_GEN: Transport unit

Event CE_AN: BOM item

Event CE_AB: Batch status

Attribute 4

Internal use

Attribute 5

Event CE_AB: Current batch

Counter 1, Type 1, Reason 1

Internal use

Counter 2, Type 2, Reason 2

Internal use

Counter 3, Type 3, Reason 3

Internal use

Counter 4-10, Type 4-10, Reason 4-10

Internal use

Partitioning

Partitioning. Is currently only displayed for the event M_TLG.

Displayed events

The following events are displayed:

Event	Designation	Type
A_AN	Log operation on	Order-related event
A_TR	Partial confirmation	Order-related event
A_UN	Interrupt operation	Order-related event
A_AB	Log operation off	Order-related event
A_BE	Terminate operation	Order-related event
A_MR	Partial confirmation	Order-related event
P_AB	Log person off	Person-related event
P_AN	Log person on	Person-related event
P_VAN	Person advance logon	Person-related event Cannot be edited
M_MST	Change workplace/ machine status	Machine-related event
M_AUTO	Automatic quantity posting from the terminal	Machine-related event

Event	Designation	Type
M_CTR_AUTO	Automatic counter posting from the terminal	Machine-related event
M_TLG	Change in partitioning	Machine-related event
M_SZY	Change target cycle	Machine-related event Cannot be edited
M_PSPERRE	Production lock (beginning with MDE 7.2)	Machine-related event Cannot be edited
CA_AB	Log output batch off (MPL)	Batch-related event
CA_AN	Log output batch on (MPL)	Batch-related event
CE_AB	Log input batch off (MPL)	Batch-related event
CE_AN	Log input batch on (MPL)	Batch-related event
NC_AB	Deactivate OP (ADE-BEA)	Miscellaneous event
NC_AN	Activate OP (ADE-BEA)	Miscellaneous event

The colors for the events listed are as follows:

Color	Meaning
Blue	Order-related events
Green	Person-related events
Red	Machine-related events
Brown	Batch-related events
Black	Locked events
Purple	If the order type option "Change after upload" is set to "Allow no changes" or "Allow modification, no upload", then all the events already uploaded to the ERP system are displayed in purple.

Each event-triggered dialog is apparent from the "Dialog" column. In addition to the dialogs of the same name (e.g. A_AN, A_UN, ...) the following dialogs, among others, can be triggers for the events described above:

Dialog	Meaning
A_P_AN	Log order + person on together (A_AN + P_AN)
P_AAB	Log all staff off (1..n P_AB)
A_AUN	OP automatically interrupted as a result of a shift change
A_AAN	OP automatically logged on again as a result of a shift change
SA_AN	Log merged operation on
SA_TR	Partial confirmation with merged operation
SA_AB SA_ABME	Log merged operation off
P_KOM	PZE Clock-in
P_GEH	PZE Clock-out
CA_WL	Output batch change

Display automatically recorded counter values

The counted values that have not yet been evaluated are recorded and posted as event M_CTR_AUTO at workplaces the automatic quantity collection of which is performed using the Windows terminal software. The counted values from the counter events of M_CTR_AUTO are not displayed in the columns for yield, scrap, etc., but rather in separate "counter" category columns.

The primary quantities resulting from the counted values (evaluated) are also logged and displayed as information in the columns for yield, scrap, etc.



The resulting (evaluated) quantities are not logged into other quantity units. That is, basic, secondary and tertiary quantities are always 0 for the counter results.

The resulting (evaluated) quantities are logged irrespective of the reason.

Recalculation of modified data



After modifications have been completed, recalculation will start by clicking on the  icon and by acknowledging the confirmation prompt. Depending on the extent of the modifications made, recalculation may take some time.

The following steps are performed during recalculation:

1. Validity check of the modified data

Here, as with online data entry, all events involved are checked to make sure they can be fully processed. If any events fail the validation check, then the recalculation is rejected without any modifications being made in the system. If a validation error occurs, then an error message and a notice referring to the event involved are displayed accordingly notifying the operator.

2. Deleting existing results

If all the validation checks were completed successfully, then the still available, current results are canceled first. If postings have already been uploaded to the ERP system (upload identifier "J"), then the corresponding cancellation records are generated in the same way postings are modified manually in posting maintenance. The quantities and durations contained in the order-related posting are also canceled in the operation status.

3. Calculating new results

After a successful validation check and cancellation, the modified events are reevaluated and a new posting is generated.

The quantities and durations calculated for order-related posting are also posted to the operation status.



However, keep in mind here that any configuration changes that have been made in the meantime may now quite possibly make it impossible to make any subsequent modifications to already entered scenarios or that this can lead to different results.

Example: If two operations were logged onto a machine at the same time, and in the meantime the option *parallel OPs can be logged on* was deactivated, then the recalculation of one operation will bring about a validation violation. Recalculation is now no longer possible.

The user is notified accordingly once recalculation has completed successfully.

Locking concept in event maintenance

Edited data is selectively locked based on the chosen machine and the time domain entered.

All locked entries in event maintenance are displayed under *System administration > Locked data records* and can only be deleted by a user who has the appropriate function authorization.

Deleting a locked entry will result in the inability to recalculate data currently displayed on that console. The user must discard the scenario and request data again.

The "parallel staff logins" option in event maintenance

If multiple machine operation is taking place, this option must be activated in the event maintenance selection area, because doing so will divide up time and labor data to match the order postings.

In the process, all machines are displayed at which the staff displayed was logged onto during the evaluation period.

If, however, the nesting depth is too great, it might not be possible to recalculate the data, because the starting events for this are no longer in the period selected.

Example:

Person 1 is logged onto workplaces 100 and 200

Person 2 is logged onto workplace 200 at a staggered overlapping time

Person 3 is logged onto workplaces 200 and 300 at a staggered overlapping time

Because the recalculation of all events must be considered for an order, and because of nesting more and more order events must be recalculated, this may bring about a constellation in which a scenario can no longer be recalculated.

Please note: Event maintenance is unusable when personnel postings are this deeply nested.

The "optimized parallel staff logins" option in event maintenance

When this option is selected, the following optimized processing takes place with parallel staff logins:

- Only data for the selection are displayed or requested (machine, order/OP, person)
- Machines that also have to be recalculated because of parallel staff logins are not added to the selection until the recalculation process.
The additional machines are determined based on the modified data.
- If an error occurs during recalculation, then the additional machines are subsequently also displayed, and the user can correct the data.

Waiting period processing

Waiting period processing is an optional processing that controls how the system functions with respect to entering personal postings. It is activated through the basic settings.

When the waiting period is exceeded, a separate waiting period posting is generated (staff postings to the defined waiting period operation). If the waiting period is not exceeded, then the staff posting is backdated (and the OP posting also, if required).

The posting times of the events remain unchanged during waiting period processing, i.e. the posting times are edited in event maintenance.

When events are recalculated, a new decision is made depending on the modified posting scenario about whether to backdate because of the waiting period processing.

Modifications after upload

Whether modifications in event maintenance should be uploaded to the ERP system is determined based on order type during the customizing process. The following options are available:

- Allow modification and upload
- Allow modification, no upload - data can be modified, but are not uploaded→
- Do not allow modification - data can no longer be modified.

The verification is based on workplace: If an order at a workplace is not modifiable according to the order type configuration (purple font color), then all events are locked that are older than the last confirmation/upload date of this order. This lock is order independent and so also affects, for example, orders that are modifiable.

→→

21 Foreman's Checklist

Summary

Menu	Information management → Messages → Foreman's checklist
Transaction code	fmchkl
Function authorization	fmchkl

Usage

The foreman's checklist is provided to allow the foreman to monitor operations produced in the foreman's area:

- It shows any out-of-the-ordinary data (e.g. time or quantity deviations)
- It shows postings that the foreman must authorize.

The information listed on the foreman checklist is generated daily (by an application started on the server automatically overnight) for the last seven days (default setting) and is stored in a database table . The application analyzes the information detected.

Integration

The data displayed here (e.g. time, quantity deviations) result from the data entry at the shop floor entry units (e.g. terminals).

Selection criteria

The application provides the following selection criteria:

Order

Restricted to order number. There is an option to select using wildcards.

Category

Restricted to order categories.

Order type

Restricted to order types.

Workplace

Restricted to workplace number; only postings are displayed that relate to the workplace number entered.

Responsibility area

Restricted to responsibility area; the postings are displayed that relate to the workplaces that are assigned to the responsibility area entered.

Cost center

Restricted to cost center; the postings are displayed that relate to the workplaces that are assigned to the cost center entered.

Company

Restricted to company; the postings are displayed that relate to the workplaces that are assigned to the company entered.

Posting

The following information is logged. By clicking on the check box *Posting* you can limit which posting categories are displayed:

Postings with target-actual quantity deviation exceeding +/- 5%

Finished operations, for which a yield (in primary quantity unit) is posted of more than +/- 5% as compared to the target quantity (in primary quantity unit).

Postings with target-actual time deviation exceeding 10%

Finished operations with an order duration (times that are posted to RPA 11) of more than 10% of the target processing time. Here, only those operations are considered that themselves have a target processing time that is greater than 0.

Non-authorized personnel postings

Non-authorized staff-related BDE log records (record type "B"). The log records can be authorized using the application [order-related postings](#).

During customizing you can define by order type whether or not these postings should be logged.

Non-authorized order postings

Non-authorized BDE log records relating to orders (record type "U", "E") for overhead cost operations. These log records can also be authorized using the application [order-related postings](#).

During customizing you can define by order type whether or not these postings should be logged.

Open operations of finished production orders

Any still active (prepared, running or interrupted) operations of production orders (Category "FA"), with a final operation already showing a status "finished".

Postings showing scrap without reason

Log-offs or interruptions for operations are displayed for which the defined scrap reason is 999.

This requires that scrap reason 999 was configured and that scrap postings were made manually at the terminal using scrap reason 999.

Show completed postings

If this checkbox is not set, only pending postings will be shown. If the checkbox is enabled, completed as well as pending postings will be shown.

In general, only data is output which the user is authorized for by the responsibility area. The responsibility area of the machine/workplace where the order was produced is referred to if data based on operations is selected. The user's responsibility area is used as selection criterion if personal data is displayed.

Field descriptions

The postings generated daily are shown in table form in the foreman's checklist. It displays not only information about the posting itself, but also the master data (workplace, order, persons) imported from the HYDRA data pool charts.

The "posting" column of the totals line shows the number of entries.

Posting category

Posting: refer to section [Selection parameters](#)

Production date/production time: the data's origin for this value depends on the type of posting:

- Postings of finished operations with target/ actual deviation: Time of most recent log-off
- Open operations belonging to the finished production orders: Time of the most recent posting (for running and interrupted operations)
- Non-authorized personnel/order postings: Log-off time of the log record
- Postings showing scrap without reason: Log-off time of the log record

Operation category

Order

Operation

Article (of the operation)

Article designation

OP designation

"Miscellaneous" category

Workplace

Short name (of the workplace)

Person (personnel number)

Name (first and last name)

Last name

First name

Order category

Evaluation date (start of the evaluation period; the start of the evaluation period is calculated as follows: "today" - INTERVAL. When calling up the hy_mst application on the server, INTERVAL can be transferred as a parameter. If this parameter is not transferred, 7 is assumed as the default value).

Category (order category)

Order type

Final article (article in the order header)

MRP controller

Order group

Project number

Sequence

Split

First logon (for the operation, date/time)

Last posting (for the operation, date/time)

Target quantity: target quantity of the operation (primary quantity unit), only filled out when the "target/actual quantity deviation" is posted.

Actual quantity: yield of the operation (primary quantity unit), only filled out when the "target/actual quantity deviation" is posted.

Target duration: target processing time of the operation, only filled out when the "target/actual time deviation" is posted.

Actual duration: main utilization time posted onto the operation (RPA 11), only filled out when the "target/actual time deviation" is posted.

Workplace category

Designation

Group

Cost center

Company

Responsibility area

Person category

Company

Cost center

Area

Department

Responsibility area

Signing category

Done

Shows whether or not the entry has already been marked "done" by a user.

Modified by

ID of the user who marked the entry as "done".

Done on

Point in time when the user marked the entry as "done".

Toolbar

Go to category



Order information

Function authorization: orin

[Order information](#) is called up. The order number is transferred as the parameter.



Order overview

Function authorization: orov

[Order overview](#) is called up. The order number is transferred as the parameter.



Workplaces/ machines

Function authorization: wpov

This calls up the application [Workplaces/ machines](#). The workplace number is transferred as the parameter.



Order related postings

Function authorization: oboo

This calls up the application [Order related postings](#). The following values are transmitted as parameters: Workplace, order, operation, production date.



Done

Function authorization: fmchkl.sign

Selected postings can be marked "done" by this function (multiple selections possible).

Processing notes

The information listed on the foreman checklist is generated daily (by an application hy_mst.exe/.out started on the server automatically overnight) for the last seven days (default setting) and is stored in a database table that is accessed by MOC.

If required, the below-mentioned call parameters can be added to the application hy_mst.exe/.out that is integrated in the HYDRA Scheduler:

/INTERVAL=days

Increase evaluation period (default: "today" - 7 days)

/DEL_UNSIGNED= days

Delete postings after x days (default: 999 days)

/NO_CERT

Personnel postings that have not been signed are not taken over into the foreman's checklist.

As part of customizing the system and as an alternative to this parameter, it can be defined for which order types

- personnel postings
- order postings

are to be signed and, as a result, shown in the foreman's checklist.